Independent reconstruction of the coastal adaptation model of Wong et al. (2025): continuing research report

Research-agent record prepared for Sophia Ludtke

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Abstract

Coastal cost distributions depend on both sea-level response and the joint uncertainty assigned to climate, exposure, and adaptation. This project examines whether the results of Wong et al. (2025) can be reconstructed from scholarly documentation without consulting the authors' implementation or computational outputs. We recover component equations, implement numerical kernels in Julia, and compare manufactured cases with analytical limits and an independent SciPy integration. Continued source recovery supplies coupled climate and Antarctic dynamics, a closed carbon-cycle alternative, and independent observational inputs. A temperature-only calibration passes convergence diagnostics but fails held-out temperature and upper-ocean heat checks. These failures motivate a separate temperature–heat calibration, which reduces held-out temperature RMSE from 0.209 to 0.087 K but underestimates assessed future warming. Independent geographic exposure, economic inputs and ice-mass observations support further explicit alternatives; the failed checks are retained. Further recovery permits paleo Antarctic hindcasts, a conditional fit to four written constraints, and emissions-driven coupled carbon–climate experiments. Complete coastal calculations at two elevation resolutions separate a numerical exposure approximation from missing target geographic specifications. The calculations verify heat conservation, ice-response convergence, flood-loss integration, and economic accounting. They also expose a finite-step residual in the printed emissions recurrence and a lower-bound discrepancy in an Antarctic temperature transform. These are findings about written equations, not evidence of defects in the inaccessible original software. A complete independent joint calibration and the target coastal input bundle were not recovered. Consequently, none of the thirteen numerical target figures is independently reproduced. An original workflow diagram, verified component calculations, explicitly labeled alternative experiments, a complete figure register, and recovery requirements document ongoing work. This report does not conclude that all scientifically feasible reproduction work is exhausted.

1 Objective and evidentiary boundary

Wong et al. [1] estimate global coastal adaptation costs and losses conditional on warming in 2100. Their model connects emissions, carbon cycling, temperature and ocean heat, global and local sea level, and segment-level coastal decisions. The scientific objective here is to reproduce the resulting conditional distributions from the documented mechanisms. It is not sufficient to draw curves resembling the published figures or to reproduce marginal parameter ranges while discarding their dependence.

The independence restriction permits scientific papers, their supplementary methods, and exogenous inputs from independent providers. Published figures and printed summary statistics are comparison evidence only. No source code, processed input bundle, posterior sample, simulation file, or plotting script from the target authors was opened or used. An initial broad literature search

displayed incidental repository-index snippets, including a data-loading excerpt. Those results were discarded, and subsequent searches focused on scientific documents and provider records. This exposure is retained in the research log rather than represented as complete isolation from search results.

The recovered sources include the target article and its ten-figure supplement, the SNEASY climate description [3], its carbon-cycle reference [4], BRICK and MimiBRICK descriptions [5, 6], the DAIS paper [7], and the CIAM mathematical supplement [8]. The December 2024 revision of the cited emissions preprint [2] was recovered through public scholarly file-version metadata. Its current DOI landing page serves a later 2025 version. Since the target citation does not pin a revision, this historical manuscript remains a version-qualified specification. Stable acquisition URLs and hashes are recorded in `docs/SOURCES.csv`.

2 Original experiment and recovered interfaces

The target source ensemble contains 100,000 states of the world, with 5,000 used for the coastal analysis. Global warming in 2100, relative to 1850–1900, defines four bins: at most 1.5 °C, above 1.5 and at most 2.5 °C, above 2.5 and at most 3.5 °C, and above 3.5 °C. Global sea level in 2100 uses a 1995–2014 reference. The baseline socioeconomic scenario is SSP4; SSP2 provides a sensitivity case. Monetary outcomes are in 2010 US dollars with a 4% annual discount rate [1].

The coastal model uses 12,148 DIVA segments and decadal calculations from 2010 through 2150. Planning intervals are 2010–2050, 2060–2100, and 2110–2150. Protection and retreat standards are associated with return periods of 1, 10,100, 1,000, and 10,000 years, alongside no adaptation. The paper specifies nine adaptation combinations. The complete 2022 scholarly supplement resolves the menu: protection at 10–10,000 years and retreat at 1–10,000 years. The nominal one-year level is annual maximum high tide, obtained for the GTSR experiment by scaling the DINAS one-year level by the GTSR/DINAS 100-year ratio. It is not the GEV quantile at probability zero.

The 2022 main methods specify first-period cost minimization, persistence of the selected strategy and return level, and later height augmentation. Actual adaptation already in place is not globally observed; the original calculation initializes an optimal adaptation state. Reconstructing this spin-up requires the same exposure and extreme-water inputs as the forward model. It cannot be inferred from a global cost total.

Figure1 identifies the implemented kernels and the missing links. These kernels can be exercised independently. They do not constitute a calibrated climate-to-cost simulation.

3 Independent mathematical reconstruction

3.1 Emissions and climate kernels

For annual emissions q_t in $\mathrm{Gt\,CO_2\,yr^{-1}}$, the target's printed equation is

$$q_t = \begin{cases} q_{t-1} + \gamma_g(\tau - t), & t \le \tau, \\ q_{t-1} - 2q_\tau\gamma_d e^{\gamma_d(t-\tau)}/[1 + e^{\gamma_d(t-\tau)}]^2, & t > \tau. \end{cases} \tag{1}$$

The implementation evaluates the second branch with negative exponential arguments to avoid overflow while preserving the algebra. The growth coefficient is interpreted in the annual recurrence's time convention; a dimensional continuous-time growth coefficient would have different units. An integer peak year is required explicitly. The emissions prior allows continuous peak years, but the treatment of q_τ for a noninteger peak was not recovered.

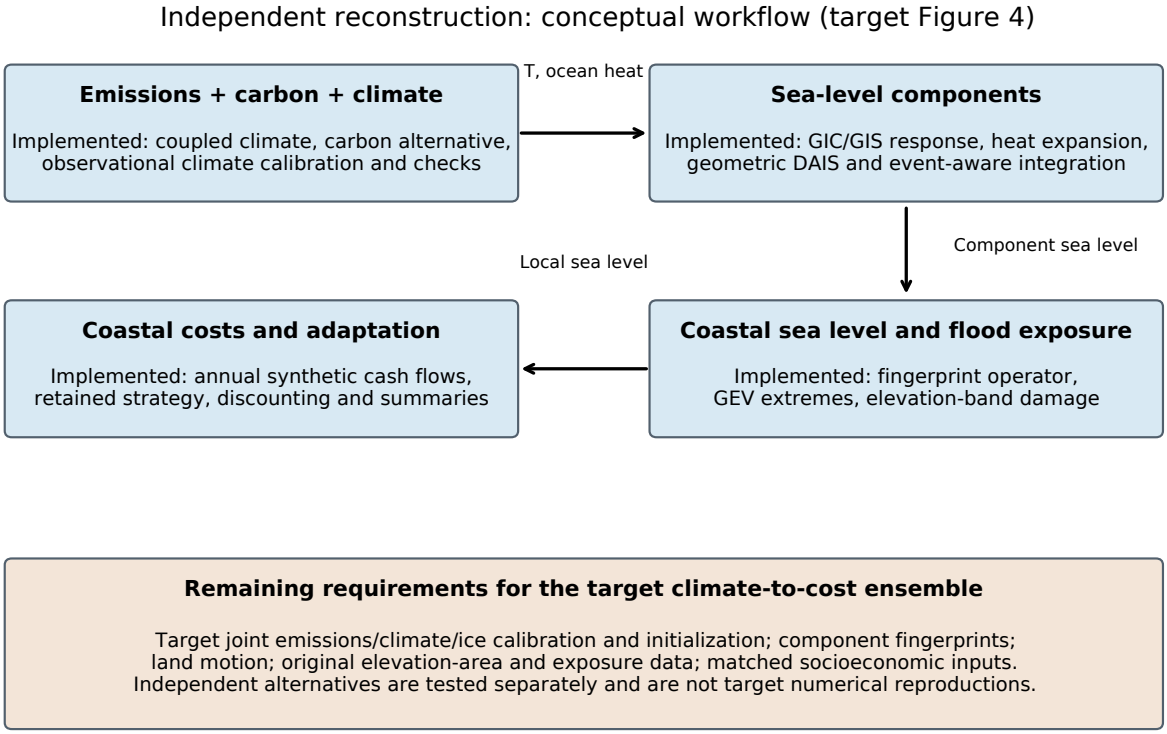


Figure 1: Original conceptual schematic corresponding to target Figure 4. Blue boxes identify implemented component calculations and their physical interfaces. The lower box lists unresolved inputs and model closure. The arrows describe required exchanges, not a claim that a complete coupled ensemble has been executed. This diagram is not counted as a reproduced numerical result.

The 2024 manuscript gives independent truncated-normal priors for growth, peak year, and decline: means $(0.004, 2070, 0.07)$, standard deviations $(0.0075, 25, 0.05)$, and bounds $[0.001, \infty)$, $[2030, 2200]$, and $[0.001, 0.2]$, respectively [2]. These are not the posterior distributions used in the target ensemble. The subsequent inversion uses expert cumulative-emissions information whose complete probability-density prescription was not recovered. Sampling these priors would therefore not reproduce the target emissions distribution.

Recoverable carbon kernels are net primary production $N = N_0[1 + \beta \ln(C/C_0)]$, with $N_0 = 60\,\mathrm{Gt\,C\,yr^{-1}}$ and $C_0 = 280\,\mathrm{ppm}$, and heterotrophic respiration $(a_1 D + a_2 S) Q_{10}^{\Delta T/10}$ [4]. Here D and S are detrital and soil carbon, and a_1, a_2 are turnover rates. The continuation recovers terrestrial allocation and carbonate equilibrium, allowing the closed, explicitly alternative calculation in Section7. The implemented CO_2 forcing is $3.7 \log_2(C/C_0)\,\mathrm{W\,m^{-2}}$ [3].

The ocean verification operator solves $\partial_t T = \kappa \partial_{zz} T$ on a one-dimensional column with prescribed surface temperature and zero bottom heat flux. Cell-centred finite volumes use half-cell distance at the upper boundary. Backward Euler time integration ensures that the column heat change equals the top flux evaluated at the new time, to roundoff. Multiplication by volumetric heat capacity converts the tested temperature integral to heat per unit area. This independently chosen discretization is not asserted to match the original SNEASY solver.

3.2 Sea-level components and spatial transformation

The glacier contribution S satisfies the documented reservoir relation

$$\frac{dS}{dt} = \beta(T - T_{\mathrm{eq}})(1 - S/V_0)^n, \qquad T_{\mathrm{eq}} = -0.15\,^{\circ}\mathrm{C}. \tag{2}$$

Temperature is referenced to 1850–1870, V_0 is initial glacier volume expressed as metres of sea-level equivalent, and β has units $\mathrm{m\,K^{-1}\,yr^{-1}}$ [5]. The numerical domain requires $S \le V_0$ for fractional n. An integration step that leaves this domain raises an error; no undocumented volume clipping is introduced.

Greenland volume V, also in metres of sea-level equivalent, follows

$$V_{\mathrm{eq}} = cT + b, \qquad \frac{dV}{dt} = (V_{\mathrm{eq}} - V)(\alpha T + \beta). \tag{3}$$

The temperature reference is 1961–1990, and the volume loss relative to its initial state yields sea-level rise. The rate $\alpha T + \beta$ must be nonnegative. Constant temperature gives an exponential relaxation and therefore an independent verification solution.

The target lineage uses a heat-driven thermal-expansion response, $\Delta S_{\mathrm{TE}} = k_H \Delta H$, not the earlier temperature-relaxation emulator [9, 2]. The coefficient k_H in $\mathrm{m\,J^{-1}}$ remains an explicit input. If the density expansion parameter α_ρ has units $\mathrm{kg\,m^{-3}\,K^{-1}}$, dimensional mass and heat accounting gives $k_H = \alpha_\rho/(AC_p\rho^2)$. This is a dimensional derivation, not recovery of a calibrated numerical coefficient. Supplying a conventional volumetric coefficient in $\mathrm{K^{-1}}$ in that expression would be a unit error.

The documented Antarctic ocean-temperature transformation is

$$T_o = T_f + \frac{x}{1 + e^{-x/a}}, \qquad x = aT + b - T_f, \qquad T_f = -1.8\,^{\circ}\mathrm{C}. \tag{4}$$

An additional threshold-triggered loss rate can be evaluated when its threshold and rate are supplied. The continuation implements DAIS accumulation, marine discharge and differentiated geometry from Shaffer[7], with sign and unit corrections derived explicitly. Present-day equilibrium and

warming tests verify these mechanisms. The target paleo calibration, initial state and full coupling remain distinct recovery tasks. Land-water increments remain externally specified.

Local relative sea level is represented by $L_i(t) = \sum_c F_{ic} S_c(t) + v_i(t - t_0)$, where F contains component fingerprints and positive v_i increases relative sea level. This sign convention includes subsidence as positive. The matrix operation is verified on known area-weighted factors; actual fingerprint grids and coast matching are absent. Reference-period routines require complete annual coverage and refuse silent rebasing of an incomplete series.

3.3 Coastal damage and costs

The independently implemented flooding calculation uses a generalized extreme value (GEV) distribution for annual extreme water level and piecewise uniform area density within supplied elevation bands [8]. For water height w and exposed elevation e, property damage is proportional to $\phi(w - e) = (w - e)/(h_0 + w - e)$ with $h_0 = 1\,\mathrm{m}$. The integral over an exposed band $[a, b]$, where $b \leq w$, is

$$\int_a^b \phi(w - e)\, de = (b - a) - h_0 \ln\left(\frac{h_0 + w - a}{h_0 + w - b}\right). \tag{5}$$

Multiplication by area density and capital density yields event loss. Mortality uses flooded population, a supplied mortality fraction and value of statistical life. Both contributions are multiplied by $1 - r$, where r denotes economic resilience. Integration of this loss against the GEV density produces expected annual loss. The surge integration bound is explicit, and the truncated density is not renormalized. Retreat changes the lowest retained elevation; protection changes the overtopping threshold.

Relocation combines income-related moving cost, mobile-capital transfer cost, and demolition of immobile capital. The documented benchmark mobile share is one quarter; immobile capital is three quarters. Reactive relocation costs five times the proactive counterpart in the target model. Separate kernels calculate inundation with explicit depreciation, wetland loss with a critical rise rate, and quadratic-height seawall construction. Annual maintenance is returned as a separate cash flow. The numerical coefficient in the construction formula must have units consistent with its quadratic height dependence; the quoted historical reference cost is not silently substituted where its unit convention is ambiguous.

For annual payments C_y, the explicit discount operator is

$$\mathrm{PV}_{y_0} = \sum_y \frac{C_y}{(1 + r)^{y - y_0}}. \tag{6}$$

A 2010 base year and 4% rate are used in verification. This base-year convention does not resolve the target caption's wording about discounting through/to 2150. The optimizer chooses the least-cost member of a supplied finite candidate menu. It does not construct the original adaptation trajectories, because their physical and socioeconomic inputs are missing.

4 Verification results

All 81 original-kernel Julia assertions pass across seven groups: emissions/carbon/forcing; ocean conservation and convergence; sea-level limits; spatial accounting and references; extreme levels and damage integration; costs and optimization; and ensemble/regional arithmetic (the sea-level group includes ODE accuracy). These original tests are numerical verification tests. Additional mechanism tests and observational checks are described in Section7. The tests require no third-party Julia

packages. Configuration values are explicitly marked as manufactured in `config/validation.toml`. Table1 is generated directly from the diagnostic files and the independent SciPy checks.

Verification quantity	Calculated value
Greenland finest absolute error (m SLE)	5.398e-10
Ocean finest maximum temperature error (K)	6.759e-06
Ocean maximum heat-balance residual (K m)	1.021e-16
Expected flood loss, SciPy reference (USD/year)	442422.479342
Expected flood loss, Julia relative difference	1.271e-11
Depth integral maximum difference (USD)	9.313e-10
Annual emissions residual at gamma=0.07 (% of peak)	1.750000
Minimum sampled Antarctic ocean transform (degree C)	-2.078464

Table 1: Computed verification metrics. These quantities describe manufactured test cases, not target global projections. The heat-balance residual is reported before multiplication by volumetric heat capacity.

4.1 Printed emissions recurrence and its limit

Figure2 compares the literal recurrence with the continuous logistic decline $q(t)/q_\tau = 2/[1 + e^{\gamma_d t}]$. The decrement in Equation1 is the right-endpoint sample of the continuous decline rate. Summing endpoint samples does not exactly integrate that rate. For small step Δt, the residual fraction is $\gamma_d \Delta t/4$ to leading order. At an annual step and decline rate $0.07\,\mathrm{yr}^{-1}$, the calculated residual is approximately 1.75% of peak emissions. The tested rates 0.01 and $0.2\,\mathrm{yr}^{-1}$ give approximately 0.25% and 5%, respectively. Refining the quadrature step reduces the residual. These are normalized mathematical cases extending 2000 years after an arbitrary peak, not calendar-year emissions projections.

The discrepancy separates two interpretations of the published specification: literal annual accumulation and a continuous logistic trajectory that vanishes asymptotically. No choice was imposed on a global simulation. The effect on temperature and costs cannot be calculated without the missing coupled model.

4.2 ODE and ocean-diffusion convergence

Figure3 tests the Greenland relaxation against its exact constant-temperature solution. The manufactured case has $T = 2\,\mathrm{K}$, $c = -0.2\,\mathrm{m\,K}^{-1}$, $b = 1\,\mathrm{m}$, $\alpha = 0.005\,\mathrm{K}^{-1}\,\mathrm{yr}^{-1}$, $\beta = 0.01\,\mathrm{yr}^{-1}$ and initial volume 1.2 m. The equilibrium is 0.6 m and the integration lasts 100 years. Fourth-order convergence is obtained when the RK4 step is halved.

The ocean case uses a 1 m column, diffusivity $0.1\,\mathrm{m}^2\,\mathrm{s}^{-1}$ and initial temperature $\sin[\pi z/(2L)]$ K. The upper temperature is zero and the bottom flux vanishes. The exact amplitude decays as $\exp[-\kappa(\pi/2L)^2 t]$. Time steps decrease with squared grid spacing, so the first-order temporal and second-order spatial errors converge together at second order in spacing. These deliberately artificial dimensions isolate the numerical operator from Earth-system calibration.

4.3 Flood-loss integration and decision sensitivity

Figure4 uses a manufactured $5\,\mathrm{km}^2$ segment distributed uniformly over 0–5 m elevation. Capital density is one million USD km^{-2}, population density $100\,\mathrm{km}^{-2}$, mortality fraction 0.01, value of life 100,000 USD and resilience 0.2. These values are verification fixtures, not recommended valuations.

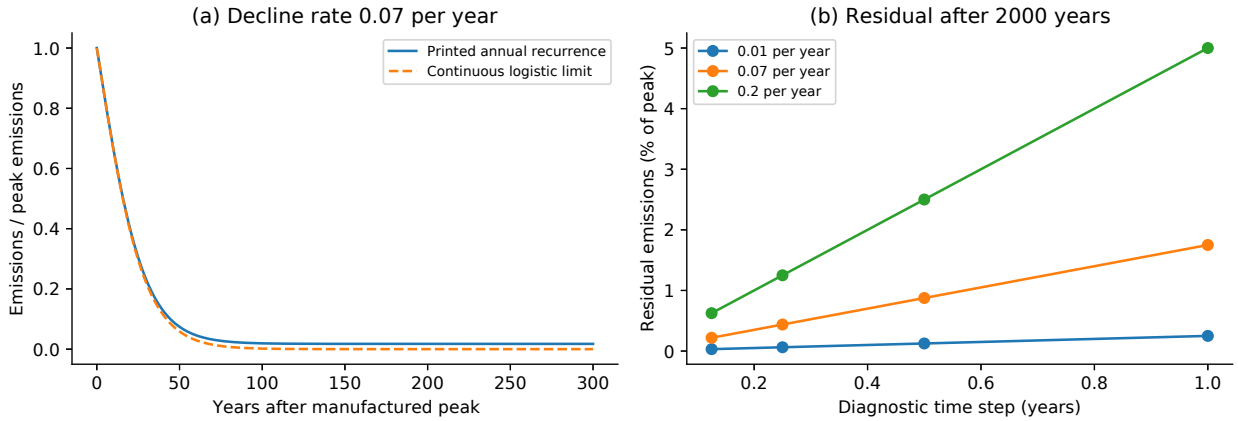


Figure 2: Independent diagnostic of the printed emissions equation. (a) Annual endpoint accumulation for normalized peak emissions and decline $0.07\,\mathrm{yr}^{-1}$, compared with the continuous logistic expression. (b) Remaining emissions after 2000 years for three decline rates and four diagnostic step sizes. The residual decreases with step size; this does not establish which interpretation the original software used.

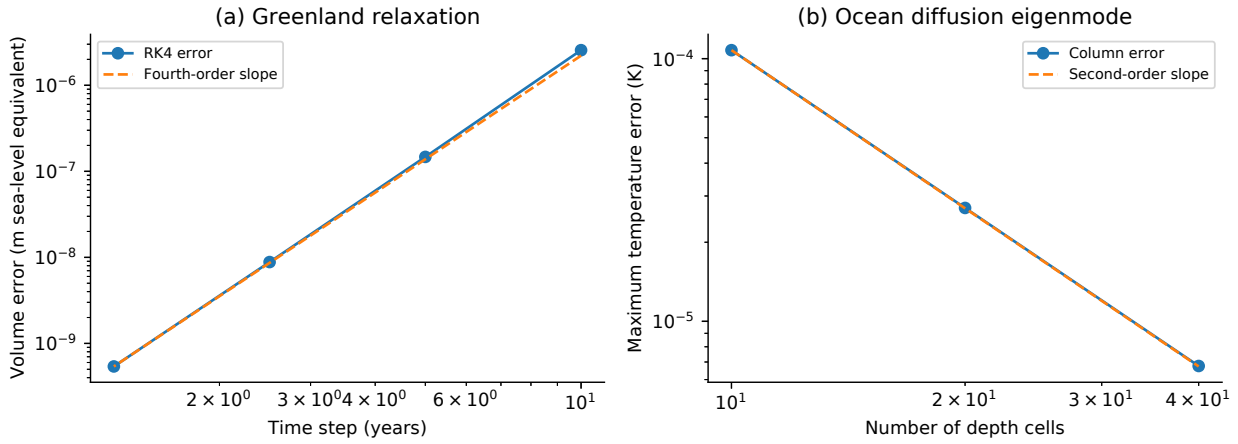


Figure 3: Independent convergence tests. (a) Greenland volume error at 100 years against the exact exponential relaxation; the dashed line has fourth-order slope. (b) Maximum ocean temperature error at 0.2 s for 10, 20 and 40 cells; the dashed line has second-order slope. Time step is 0.2 divided by the square of the cell count, in seconds.

The GEV parameters are location 1 m, scale 0.3 m and shape 0.1. The outer integral extends from −2 to 20 m. SciPy uses its opposite-sign shape convention and independent adaptive integration for both elevation and surge. This verifies the Julia closed-form depth integral and composite-Simpson expectation.

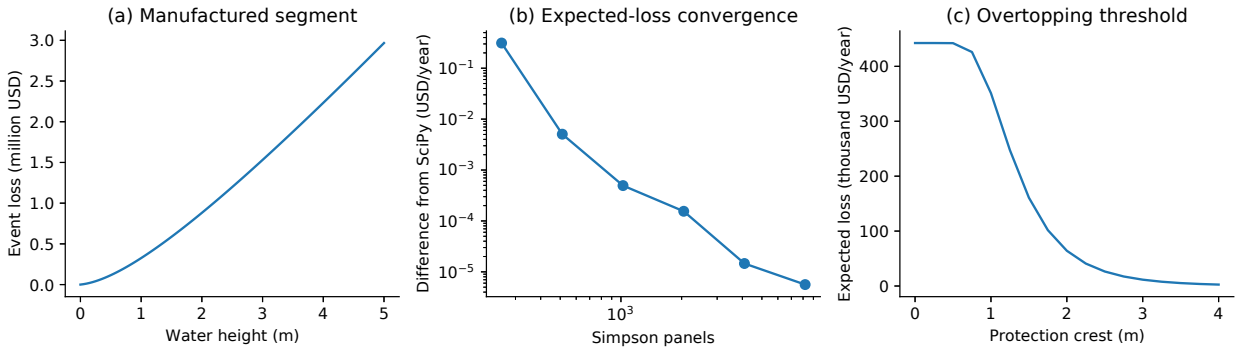


Figure 4: Manufactured coastal damage verification. (a) Event damage increases with water height. (b) Expected annual loss approaches an independent nested SciPy integral as the number of Simpson panels increases. (c) Expected overtopping loss decreases with protection crest. Panel(c) omits construction cost and therefore is not an optimal seawall-height calculation.

Table2 quantifies two accounting ambiguities with manufactured costs. A two-period menu has candidate costs $(1, 9)$ and $(6, 2)$ at years 2010 and 2060. Keeping one candidate differs from allowing a switch. This menu has no inherited infrastructure state; it is a test of the decision rule only. Likewise, ten annual unit payments during 2010–2019 differ from paying ten units at either end. A coarse timestep must retain the annual-flow interpretation if it is to reproduce the same present value.

Manufactured accounting choice	Present value
fixed strategy pv	2.266414
revisable strategy pv	1.281425
ten annual unit payments pv	8.435332
ten units paid at start pv	10.000000
ten units paid at end pv	7.025867

Table 2: Sensitivity to strategy persistence and payment timing at 4% annual discounting. Units are arbitrary currency; these are not coastal estimates.

4.4 Antarctic transform and its asserted floor

For $a > 0$ and $x < 0$, the added term in Equation4 is negative. The literal transformation therefore falls below T_f, despite its description as bounded below by freezing [5]. For the test values $a = 1, b = 0$, the minimum is approximately $-2.07846\,°\mathrm{C}$. In dimensionless form, $z = x/a$, the extremum satisfies $1 + e^{-z} + ze^{-z} = 0$, giving $z \simeq -1.27846$ and a minimum increment $-0.27846a$. Figure5 shows this algebraic behavior. No freezing clamp was added. Whether the calibrated historical trajectory visits this regime remains unknown.

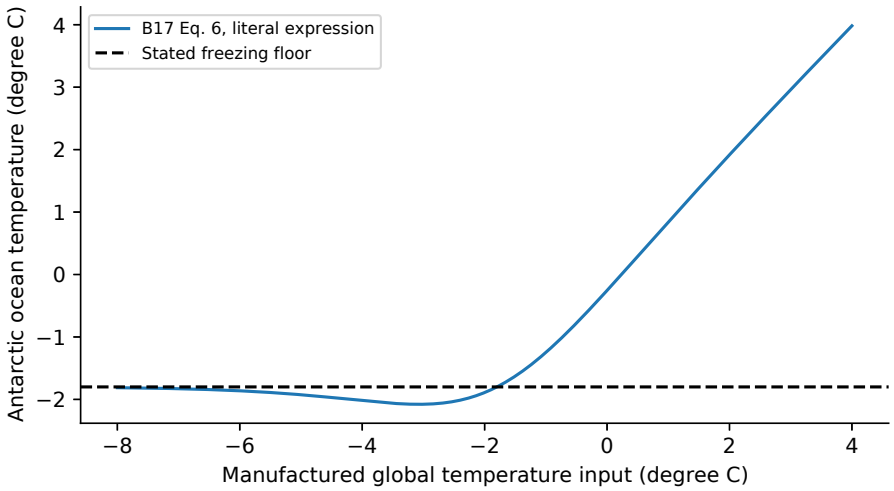


Figure 5: Literal BRICK Antarctic ocean-temperature transformation with manufactured coefficients $a = 1, b = 0$ and stated freezing temperature $-1.8\,^{\circ}\mathrm{C}$. The curve falls below the dashed floor for negative x. This is a specification diagnostic, not an Antarctic projection.

5 Quantitative comparison boundary

The target reports median costs of 1.2, 1.5, 2.1 and 2.6 trillion 2010 USD across the four warming bins, with fifth percentiles 1.0, 1.2, 1.6 and 2.2 trillion USD [1]. The prose supplies 95th percentiles 1.3 and 2.3 trillion USD for the first two bins; unreported numerical endpoints were not digitized or invented. No independent global costs exist here, so neither relative errors nor agreement intervals can be calculated for those targets.

An arithmetic audit of the printed Table 1 is possible. Its four counts sum to 5,000. Among 250 simulations above 2.7 trillion USD, the text gives 125 in the highest bin and 109 in the third, with none in the lowest. The remaining 16 belong to the second bin. Dividing these counts by bin size yields the rounded probabilities in the paper (Table3). This verifies internal consistency of printed summaries; it does not reproduce the modeled probability distribution or establish an independently calculated tail threshold.

Bin	Printed count	Tail count	Calculated%	Printed%
1	202	0	0.0000	0
2	2264	16	0.7067	1
3	2145	109	5.0816	5
4	389	125	32.1337	32

Table 3: Arithmetic audit using published counts only. The second-bin tail count 16 is derived by subtraction. These data never enter the model kernels.

6 Complete figure-by-figure record

Table4 covers all main and supplementary figures and the main table. Unlettered joint and marginal views are listed separately. Every entry's dependencies, source provenance, assumptions, commands,

comparison status and failure explanation are also recorded in `docs/FIGURE_STATUS.csv`. For unavailable outputs, the generation command is explicitly unavailable; there is no plotting command that can turn missing model predictions into an independent reproduction. Supplementary Figure 10 is one histogram containing all-member and Antarctic-trigger-conditioned distributions of proactive migrants.

Table 4: Target reproduction status. "Not reproduced" means no independently generated target numerical result.

Figure/view	Quantity	Status
1: global interval plot	Conditional median and 5th-95th cost quantiles	Not reproduced
2: joint scatter	Sea level 2100 versus discounted costs	Not reproduced
2: top marginal	Sea level distribution by warming bin	Not reproduced
2: right marginal	Cost distribution by warming bin	Not reproduced
3: regional bars	Regional cost/GDP minus global cost/GDP	Not reproduced
4: workflow	Model components and exchanges	Conceptual schematic
S1: interval plot	5000 versus 4000 subsample comparison	Not reproduced
S2: interval plot	202 per bin versus full sample	Not reproduced
S3: trajectory plot	GMST median and 90% intervals plus AR6 boxes	Not reproduced
S4: trajectory plot	GMSL median and 66% intervals plus AR6 boxes	Not reproduced
S5: map	Median local sea level 2100	Not reproduced
S6: interval plot	GTSR versus DINAS cost intervals	Not reproduced
S7: joint scatter	Sea level versus costs above 2.7 trillion	Not reproduced
S7: top marginal	Upper-tail sea level distribution	Not reproduced
S7: right marginal	Upper-tail cost distribution	Not reproduced
S8: regional bars	SSP2 regional cost/GDP deviations	Not reproduced
S9: regression plot	Regional SSP income contrast versus cost sensitivity	Not reproduced
S10: histograms	Migration conditional on AIS trigger	Not reproduced
Table1: table	Warming-bin counts and tail probabilities	Not reproduced

Main Figure 1 requires full conditional cost distributions. Figure 2 requires joint sea-level/cost realizations, and Figure 3 additionally requires regional economic aggregation. Supplementary Figures 1 and 2 require controlled subsampling of a legitimate ensemble, not merely a sample-size calculation. Figures 3 and 4 require calibrated time series plus AR6 comparisons. Figure 5 requires spatial fingerprints and site matching. Figure 6 requires matched GTSR and DINAS extreme levels. Figure 7 requires the original type of upper-tail joint distribution. Figures 8 and 9 require paired SSP2/SSP4 economics. Figure 10 requires retained segment migration histories and Antarctic-trigger states. The implemented summary and aggregation routines do not supply these missing physical inputs.

7 Continued recovery, implementation and observational experiments

The earlier assessment identified missing mechanisms too broadly. Reading the complete Tanaka thesis and the 2022 scholarly preprint recovered the DOECLIM constants, the coastal decision menu and the socioeconomic priors. Independent providers supplied forcing, temperature, carbon emissions, ocean heat and coastal-hazard inputs [18]. The active audit in `docs/BLOCKER_AUDIT.md` distinguishes recovered specifications, derived corrections, explicit alternatives and remaining data gaps.

7.1 Coupled energy balance and numerical heat transport

The land–troposphere and ocean–troposphere boxes exchange heat and lose energy through separate radiative feedbacks [14, 3]. For land fraction $f = 0.29$, their temperatures T_L, T_S satisfy

$$C_L \dot{T}_L = Q - \lambda_L T_L - \frac{k}{f}(T_L - bT_S), \tag{7}$$

$$C_S \dot{T}_S = Q - \lambda_S T_S + \frac{k}{1-f}(T_L - bT_S) - F_O. \tag{8}$$

Here $b = 1.3$, $C_L = 0.52$ and $C_S = 7.8\,\mathrm{W\,yr\,m^{-2}\,K^{-1}}$. The global surface-air observable is $fT_L + (1-f)bT_S$. The equilibrium land/SST ratio is 1.43 and $k = 1.59 - 0.31\lambda_L$. Together with an equilibrium warming of ECS under $3.7\,\mathrm{W\,m^{-2}}$, these relations determine both feedbacks and the exchange coefficient. The explicit formulas are recorded in `docs/RECOVERED_SPECIFICATIONS.md`.

An 80-cell, 4000 m interior ocean exchanges heat continuously with the mixed layer and has an insulated bottom. Finite-volume fluxes cancel under a global heat-capacity weighted sum; this is tested independently of equilibrium fitting. Crank–Nicolson integration uses four steps per year. Refining the vertical grid from 80 to 160 cells changes the year-70 warming under 1% annual CO_2 growth by 0.000787 K; the next doubling changes it by 0.000197 K. This approximately fourfold error reduction agrees with second-order spatial convergence.

The 60 m effective mixed-layer capacity contains about 2.5 m of atmospheric-equivalent capacity. Comparisons with NOAA integrate only 57.5 m of water and the appropriate fractional interior cells down to the stated observational depth. Whole-column heat is not compared to a 700 m record. HadCRUT4 blends land air temperature with SST, whereas the legacy DOECLIM observable includes enhanced marine-air warming. Treating that blended record as the global-air observable follows the simplified legacy calibration and is an additional representation approximation, not an exact observation operator.

7.2 A closed, explicitly alternative carbon cycle

The recovered terrestrial stocks are 100, 500, 120 and 1500 Gt C in leaves, wood, detritus and soil [15]. Baseline NPP of $60\,\mathrm{Gt\,C\,yr^{-1}}$ allocates 35 to leaves and 25 to wood. Leaf turnover sends 35 to detritus; wood sends 20 to detritus and 5 to soil. Detritus respires 45 and transfers 10 to soil; soil respires 15. Linear turnover rates are these fluxes divided by their source stocks. Fertilization modifies NPP and the documented Q_{10} factor modifies the two respiration fluxes [4]. All internal transfers cancel in the global carbon balance.

Ocean carbon uses an instantaneously equilibrated atmosphere–mixed-layer reservoir and three deeper reservoirs. Constant alkalinity, total boron and carbonate equilibrium determine atmospheric CO_2 and mixed-layer carbon from their combined inventory [16]. The nonlinear scalar root is solved by bisection in log hydrogen concentration. At 280 ppm the reconstructed DIC is $0.002080909\,\mathrm{mol\,L^{-1}}$, consistent with the printed H99 value to its reported precision. The mixed-layer depth and deep exchange parameters are from the later Tanaka description. Mixing these documented versions, and holding carbonate chemistry temperature fixed, defines an alternative closure; it does not establish exact recovery of the target SNEASY version.

Second-order splitting couples carbon RK4 steps to climate CN steps. Twenty-six checks establish baseline equilibrium, chemical partition, stock positivity, second-order coupled convergence and total-carbon conservation. Across 18 historical sensitivities, the largest carbon-budget residual is below 10^{-11} Gt C. Conservation alone does not validate the response: the uncalibrated default predicts 461.38 ppm in 2022, compared with NOAA's 417.08 ppm. The GCB historical table has no

pre-1850 land-use emissions; the input retains blanks, while separate calculations prescribe zero or the 1850 rate before 1850. These are alternatives, not recovered observations.

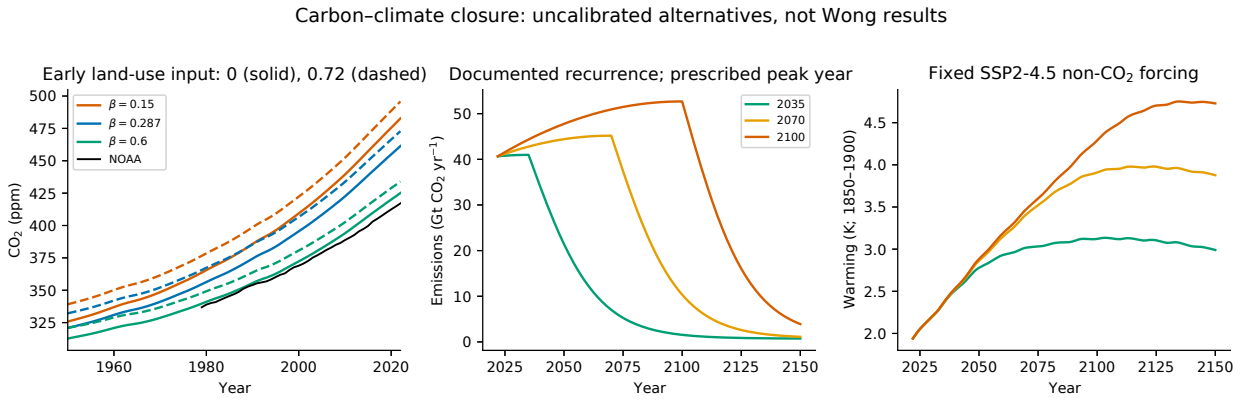


Figure 6: Uncalibrated carbon–climate experiments, not target reproductions. Left: historical CO_2 sensitivities to fertilization and the explicitly assumed early land-use input, with independent NOAA observations. Centre: documented emissions recurrence initialized from GCB2023, with three prescribed peak years. Right: coupled warming under fixed SSP2-4.5 non-CO_2 forcing. The historical mismatch limits interpretation of these future curves.

7.3 Temperature-only calibration and failed holdout checks

The first observational experiment estimates ECS, heat diffusivity, aerosol multiplier, temperature offset, AR1 innovation scale and AR1 correlation. HadCRUT4.6.0.0 years 1850–2005 form the training data; years 2006–2021 are held out. Known observational standard deviations are approximated by the total 95% interval widths divided by 2(1.96). This approximation omits the record's full temporally correlated measurement covariance. A scalar Kalman filter integrates the latent AR1 residual exactly under these assumptions; tests match the dense Gaussian covariance likelihood to 10^{-12}.

Uniform bounds follow the B17 climate table, except that ECS is restricted to 0.7–10 K: the recovered exchange relation becomes negative below about 0.68 K. CEDA effective forcing supplies historical and future inputs, with both aerosol components multiplied by the same parameter. This differs from the older indirect-aerosol convention and from Darnell's uncertain emissions and nearest-SSP forcing construction. Neither the posterior nor the six scenario experiments is claimed as the Wong ensemble.

An initial short run failed convergence checks. The retained calculation uses four dispersed chains, each with 6000 warmup and 30,000 production steps; every fifth state is retained. Proposal adaptation stops before production. Log ECS and log diffusivity proposals include their Jacobian, preserving the physical-coordinate prior. Independently implemented rank/folded split $\widehat{R}$ and bulk/tail ESS follow Vehtari et al. [17]. The largest $\widehat{R}$ is 1.00427 and the smallest ESS is 1894, satisfying the prespecified limits of 1.01 and 400.

Conditional forecasts propagate the final training-period residual state and include both AR1 innovation and observational uncertainty. Despite sampler convergence, holdout temperature RMSE is 0.209 K, with mean bias +0.189 K. Only 10 of 16 held-out annual temperatures lie inside the nominal 90% predictive intervals. After fitting only an additive heat datum offset over 1955–2005, the 700 m heat parameter-median RMSE in 2006–2021 is 16.46×10^{22} J. These failures motivate a

separate temperature–heat calibration; they are not removed from the record.

One thousand complete joint parameter rows generate six SSP trajectories; 250- and 500-row subsets test sampling variation. The physical temperature intervals exclude future residual noise. SSP2-4.5 mean warming over 2081–2100 is 2.924 K, with a parameter 90% interval of 1.976–4.188 K. The corresponding printed AR6 assessment is 2.7 [2.1, 3.5] K. This comparison uses the same averaging/reference periods but compares different inference systems; it is not a numerical reproduction of Supplementary Figure S3.

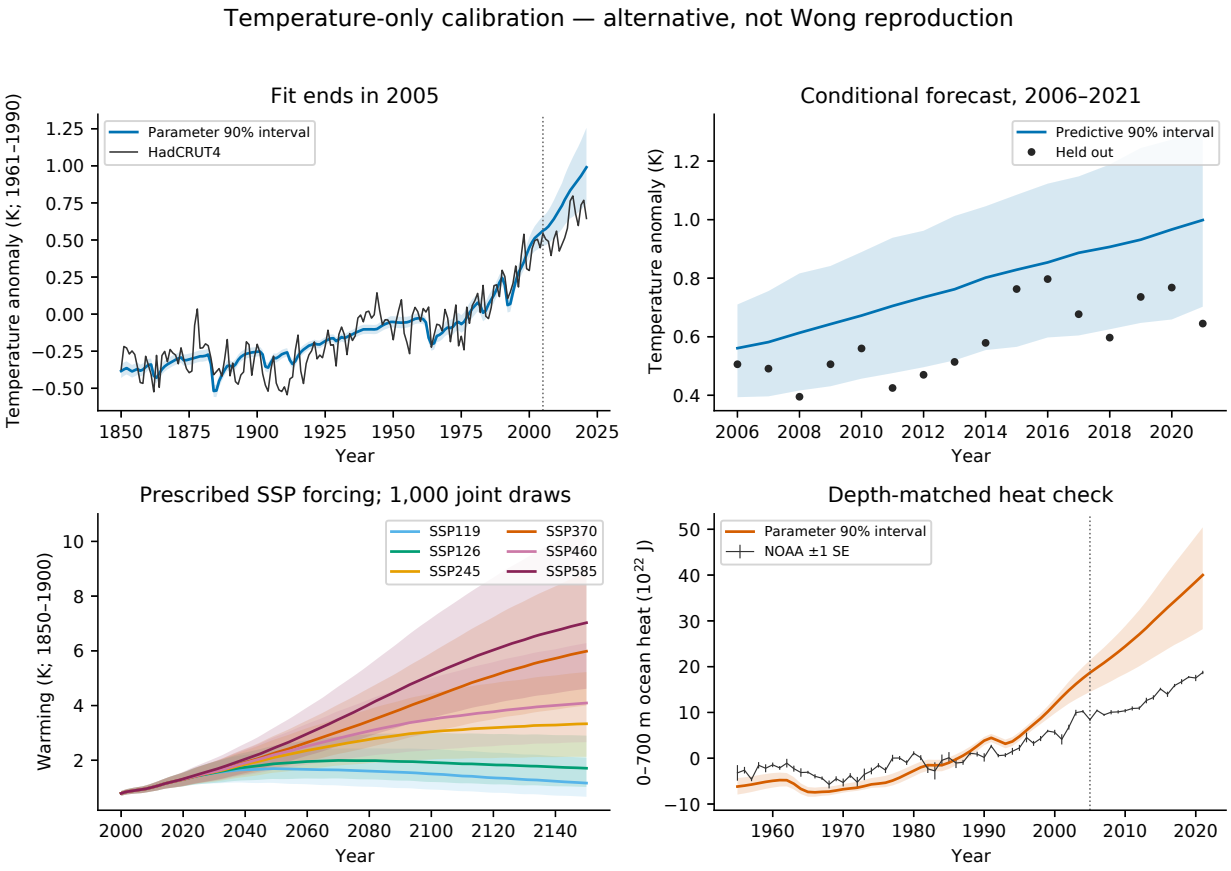


Figure 7: Temperature-only calibration and independent observational checks. The upper-right forecast includes residual and measurement uncertainty. The lower-left scenario bands contain parameter uncertainty only. Ocean heat is compared at the actual observational depth, with a training-period datum offset. The temperature and heat holdout discrepancies prevent treating this converged alternative posterior as validation of the target model.

7.4 Joint temperature–heat calibration

The next experiment adds independent NOAA 0–700 m heat observations over 1955–2005, with a datum offset and separate AR1 residual parameters. The first joint run failed convergence ($\widehat{R} = 1.0218$ and a heat-correlation ESS of about 285). Its diagnostics remain in the repository. A revised sampler uses logarithmic innovation scales, logit correlations and a recent warmup covariance window; Jacobians preserve the same physical priors. Four chains of 8000 warmup and 50,000 production iterations retain 20,000 states. All acceptance checks pass: the maximum $\widehat{R}$ is 1.00582

and the minimum bulk/tail ESS is 589.

An exact modal solver reduces computation while retaining all finite-volume degrees of freedom. A diagonal similarity transforms the climate operator into a symmetric tridiagonal matrix. Each mode is integrated analytically under linearly interpolated annual forcing. Eighteen checks agree with an independent augmented matrix exponential and establish second-order Crank–Nicolson convergence. Four substeps per year differ from the exact solution by at most 0.00133 K in three historical comparisons; the exact solver is used consistently for the new joint fit and projections.

The joint fit reduces held-out temperature RMSE to 0.0870 K and bias to +0.0402 K. Heat predictive RMSE is 0.879×10^{22} J. All 16 held-out years lie in both nominal 90% predictive intervals. These short, correlated holdouts provide limited validation; coverage is not a proof of correct uncertainty. ECS has a median of 1.485 K and a parameter 90% interval of 1.093–1.958 K. Future warming is substantially below the AR6 assessment: SSP2-4.5 warming over 2081–2100 is 1.973 [1.518,2.483] K, compared with 2.7 [2.1,3.5] K; SSP5-8.5 is 3.319 [2.569,4.160] K, compared with 4.4 [3.3,5.7] K. This discrepancy remains unresolved. Improved historical prediction does not justify replacing the target's joint climate–carbon–ice calibration with this alternative.

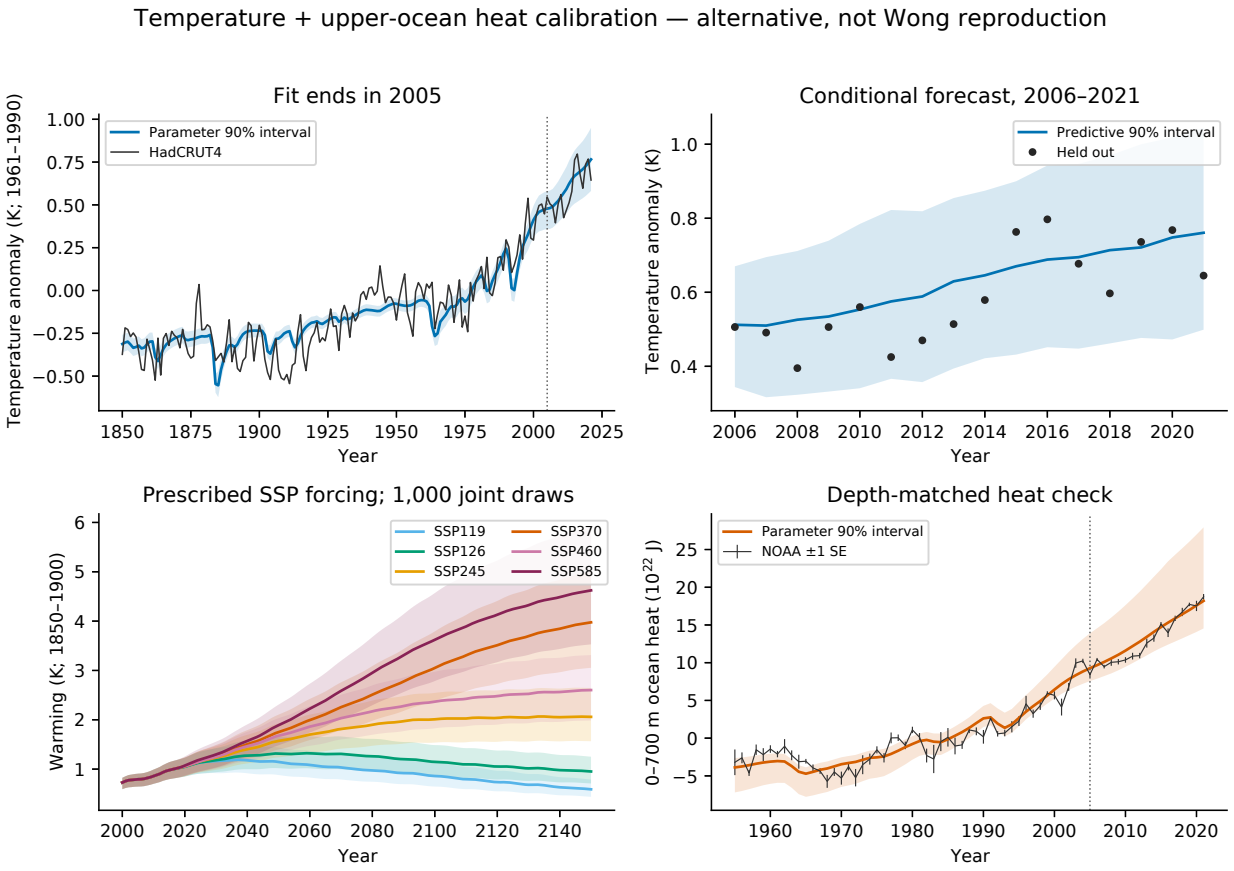


Figure 8: Alternative joint temperature and upper-ocean-heat calibration. Observations after 2005 are withheld. Future bands propagate parameter uncertainty under prescribed SSP forcing. The improved holdout performance and the low late-century warming relative to AR6 are both retained.

7.5 Antarctic events and observational thermal expansion

DAIS geometry, accumulation and marine discharge are now implemented. The discharge water depth follows sea level minus bed elevation, correcting the printed sign from geometry; coefficient units follow dimensional consistency. The equilibrium volume of $2.478479 \times 10^{16}\,\mathrm{m}^3$ differs from Shaffer's rounded $2.478 \times 10^{16}\,\mathrm{m}^3$ by 0.0193%. This is a reproduced methodological benchmark, not a reproduced Wong figure.

The printed runoff gate activates at $h_0 + cT_A = 0$. Over marine bed this can produce a discontinuous accumulation rate. Splitting time steps at that event and evaluating endpoint stages on the appropriate side removes the earlier irregular refinement. In the 6 K, century-long warming ramp, the event occurs at year 41.92982456. Two-year and one-year steps differ in year-200 radius by $1.57 \times 10^{-7}\,\mathrm{m}$; finer steps reach floating-point limits. The dashed curves in Figure9 instead activate runoff when the runoff line exceeds local edge height, and remain explicit alternative physics. These tests express sea level by mass conservation using ice/water densities and ocean area, whereas B17 scales its initial Antarctic volume to 57 m. The two observable conventions are not silently interchanged.

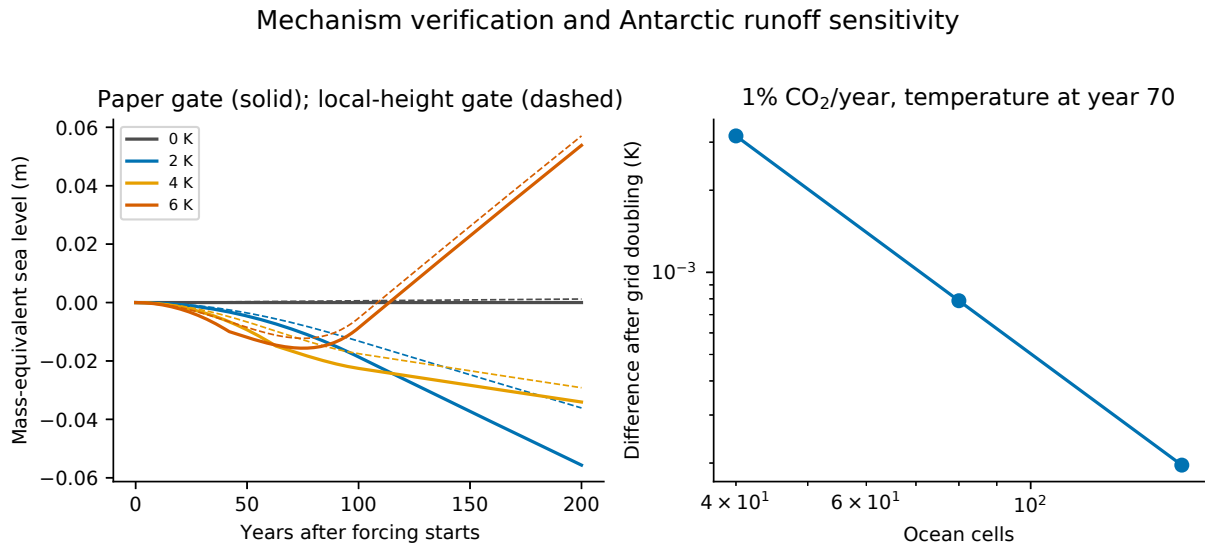


Figure 9: Independent mechanism tests. Left: prescribed Antarctic warming, with fixed ocean temperature and sea level, isolates the runoff convention. Right: climate grid refinement under 1% annual CO_2 growth. Neither panel is a global calibrated projection.

An independent coefficient check uses NOAA heat and thermosteric sea level from the same depth range. OLS over 1955–2005 at 0–700 m gives $1.392 \times 10^{-25}\,\mathrm{m\,J^{-1}}$; fitting 2005–2015 at 0–2000 m gives $1.107 \times 10^{-25}\,\mathrm{m\,J^{-1}}$. Held-out thermosteric RMSEs through 2021 are 0.414 and 0.757 mm, respectively. First-difference slopes provide a check against merely fitting a common trend. Because heat and sea level share hydrographic observations, these are descriptive constraints rather than independent-error coefficient posteriors. They are not substituted silently for the unavailable V19 prior or a full-depth expansion coefficient.

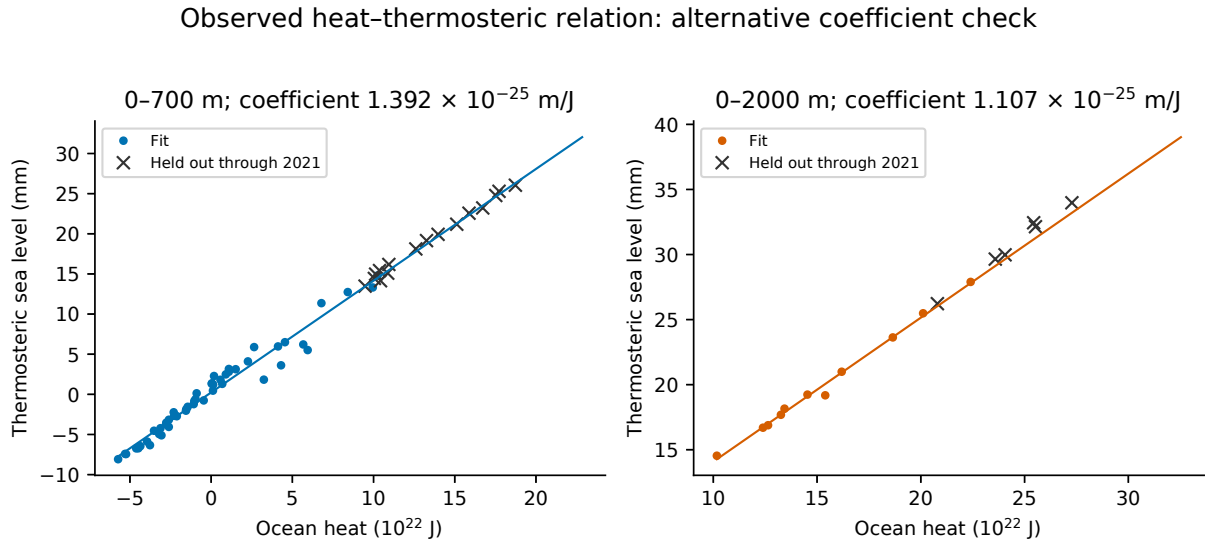


Figure 10: Independent NOAA heat and thermosteric observations with fitted depth-specific linear relations. Later years through 2021 were withheld. This coefficient check is an explicit alternative to the target calibration.

7.6 Coastal cash flows and input quality

The annual coastal experiment retains the documented ten-choice menu and first-period decision rule. Sixty checks establish nonnegative cash-flow components, first-period policy selection, no repeated relocation at fixed sea level and telescoping incremental dike construction costs. Upgrades at 2010, 2051 and 2101, annual payment timing and a paired no-SLR counterfactual are explicit alternatives to the target's decadal accounting and initial-step subtraction. Forty-eight synthetic scenarios compare income/exposure, sea-level rise, retreat land accounting and linear versus quadratic maintenance. All exposure values are labeled fixtures; no global cost estimate is inferred. Doubling flood quadrature from 512 to 1024 panels changes the verification-case NPV by about 0.00024 USD and leaves its policy unchanged.

The independent GTSR shapefile has 12,148 coastal feature rows but only 7868 distinct station labels. The initial interpretation of these labels as unique DIVA identifiers was corrected by this audit. A second public release provides 16,611 station-wise Gumbel fits. Geographic mapping leaves 133 nearest matches with invalid parameters; these remain missing, not filled from distant stations. The two input products also disagree in their 100-year levels. These findings prevent a silent merge or assumption of exact input equivalence. The full target elevation-area, exposure and DINAS tide records still require recovery. The next section uses a separately sourced geographic input product for an explicit alternative, with different segmentation and monetary conventions.

Every alternative here has its own input, script, output and figure record. No alternative curve is entered into the target inventory as a recovered published numerical result. Ongoing calibration and data-recovery tasks remain active rather than being reclassified as irreducible merely because the first implementation lacked them.

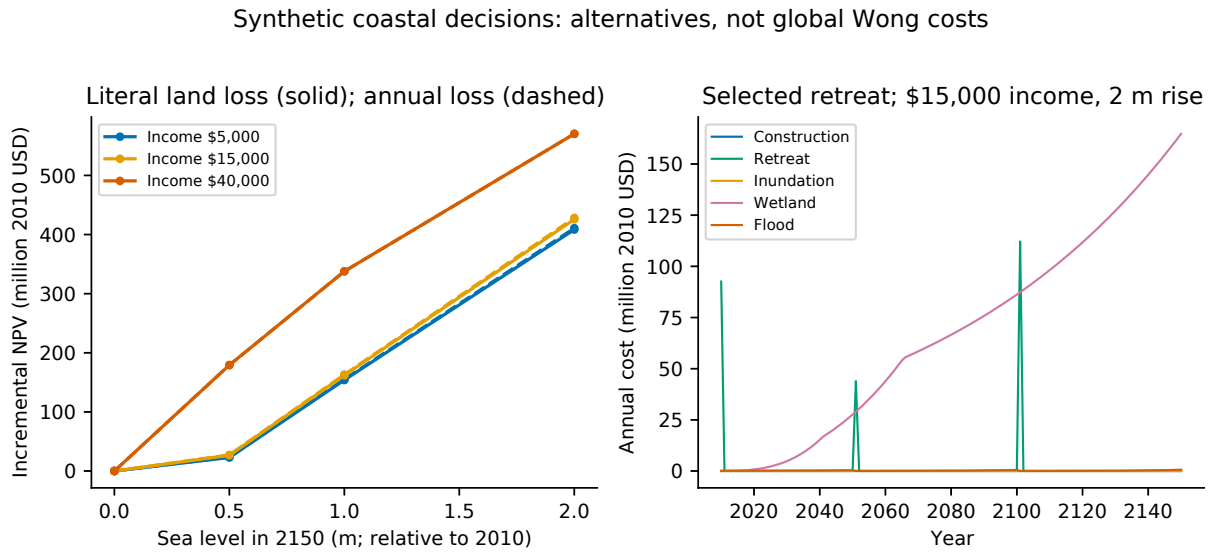


Figure 11: Synthetic coastal alternatives. Left: paired-counterfactual incremental costs under two explicit land-accounting rules. Income and exposure both vary across the three fixtures, so this is not an isolated income effect. Right: annual components for one selected retreat policy, including discrete upgrades. These outputs do not estimate a named coast or a global cost.

8 Conditional ice and geographic coastal alternatives

8.1 Carbon constraints beyond atmospheric concentration

Atmospheric concentration alone does not identify terrestrial fertilization and ocean transport separately. We prescribe the observed temperature history and fit fertilization against NOAA CO_2 in 1979–2005, retaining 2006–2021 as a holdout. The nine conditional fits vary respiration $Q_{10} = 1, 2, 3$ and mixed-layer exchange $\eta = 10, 16.87, 30\,\mathrm{m\,yr^{-1}}$. A bounded scalar search over $\beta \in [0, 2]$ is a new numerical experiment, not the wide R08 Bayesian prior. Every fit begins in 1850 at the alternative closure's 280 ppm equilibrium. This avoids the missing pre-1850 land-use series but omits earlier perturbations.

Figure 12 shows that fits with similar atmospheric residuals predict different ocean inventories. At $Q_{10} = 2$ and $\eta = 16.87\,\mathrm{m\,yr^{-1}}$, the fitted $\beta = 0.6680$ yields a 0.535 ppm holdout RMSE and 118.06 Gt C ocean uptake by 1994. R08 cites the independent estimate 118 ± 19 Gt C; its nominal 1800–1994 interval differs from our 1850 initialization, as in R08's equilibrium approximation. Predicted 1980s and 1990s ocean fluxes are 1.98 and $2.32\,\mathrm{Gt\,C\,yr^{-1}}$, compared with the written constraints 1.6 ± 0.4 and 2.0 ± 0.4. These checks do not condition the fit. Across all nine cases, time-step doubling changes CO_2 by at most 0.000333 ppm and carbon accounting closes within 7×10^{-12} Gt C. The remaining chemistry-version and prior uncertainties prevent interpreting these conditional fits as the target carbon posterior.

8.2 Independent ice observations and identifiable parameters

Independent IMBIE and GlaMBIE observations allow a conditional test of the documented Greenland and glacier equations[19, 20]. IMBIE monthly mass-change rates are averaged annually. Their errors are temporally dependent; we use the arithmetic mean of monthly standard deviations as an upper

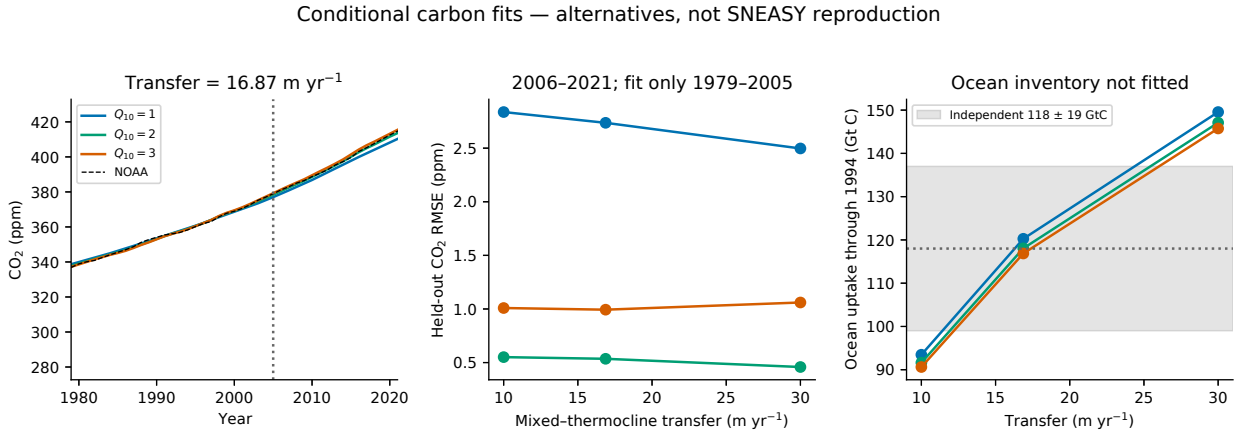


Figure 12: Conditional carbon alternatives. Left: fitted concentration histories for three respiration choices at a fixed transfer rate; the vertical line ends the fit. Centre: held-out concentration errors. Right: independently predicted ocean inventory, with the observational interval cited by R08. The latter was not used in optimization. These fits are not target-model reproductions.

bound on annual-mean uncertainty, rather than dividing by $\sqrt{12}$. GlaMBIE supplies annual rates and standard errors directly. All rates remain in $\mathrm{Gt\,yr^{-1}}$ until the explicit conversion 362500 Gt per metre of sea-level equivalent. Observed HadCRUT4 temperature is prescribed, so held-out ice years are a conditional hindcast, not a forecast made without knowledge of temperature.

The glacier equation is separable. For $n \neq 1$, define $u = (1 - S/V)^{1-n}$. Then

$$u(t + \Delta t) = u(t) - \frac{(1-n)\beta}{V} \int_t^{t+\Delta t} [T(s) + 0.15]\,\mathrm{d}s. \tag{9}$$

The integral is exact for linearly interpolated annual temperature. For $n = 1$ the remaining volume decays exponentially with the same temperature integral. Zero remaining volume is absorbing, including extinction before the end of an interval whose temperature later falls. This explicit convention resolves the nonunique mathematical continuation for $n < 1$.

For Greenland, write sea-level contribution $S = V_0 - V$ and use the documented linear equilibrium volume. The equation becomes

$$\dot{S} = [d + kT - S]10^{-3}(aT + b), \qquad d = V_0 - b_{\mathrm{eq}}, \quad k = -c. \tag{10}$$

Annual rates cannot distinguish V_0 and b_{eq} separately. Their B17 independent uniform priors induce a trapezoid density for d, which we derive from the overlap of the two intervals. We retain that density instead of inventing a uniform prior on the difference. Negative inverse response times are excluded on the modeled temperature domain. Four RK4 steps per year agree with refined and analytical cases.

We fit rates through 2014 and reserve 2015–2020. Glacier initialization in 1990 and Greenland initialization in 1961 are explicit alternatives. Physical bounds follow B17; residual innovation standard deviation is newly assigned $[0.1, 250]\,\mathrm{Gt\,yr^{-1}}$ and AR1 correlation $[0, 0.999]$. These rate-error priors differ from the original cumulative-state calibration. The first chains failed the tail effective-sample criterion and were retained as failed attempts. Only chains passing the recorded diagnostics enter subsequent projections.

The longer runs pass: the maximum $\widehat{R}$ is 1.0012 and the minimum tail effective sample size exceeds 1100. Conditional holdout RMSE is $84.8\,\mathrm{Gt\,yr^{-1}}$ for glaciers and $128.9\,\mathrm{Gt\,yr^{-1}}$ for Greenland.

Both 90% predictive intervals cover all six held-out years, but that short record does not establish predictive accuracy. A straight-line glacier trend fitted only to training data has lower holdout RMSE, 73.6 Gt yr^{-1}. The glacier model therefore does not improve on this simple baseline despite its physical structure and converged sampler. Greenland improves on the tested training-mean, last-year and linear-trend baselines (144–161 Gt yr^{-1}).

Future component calculations pair complete ice-parameter rows with complete joint climate rows independently. They do not recover the target's cross-model posterior correlations. Integration restarts from the observed-temperature hindcast state in 2021; its boundary temperature is observed, followed by modeled temperatures on the same reference datum. Future ribbons include parameter uncertainty only. The 0–2000 m thermal component uses the independently fitted NOAA coefficient, excluding unobserved deeper expansion.

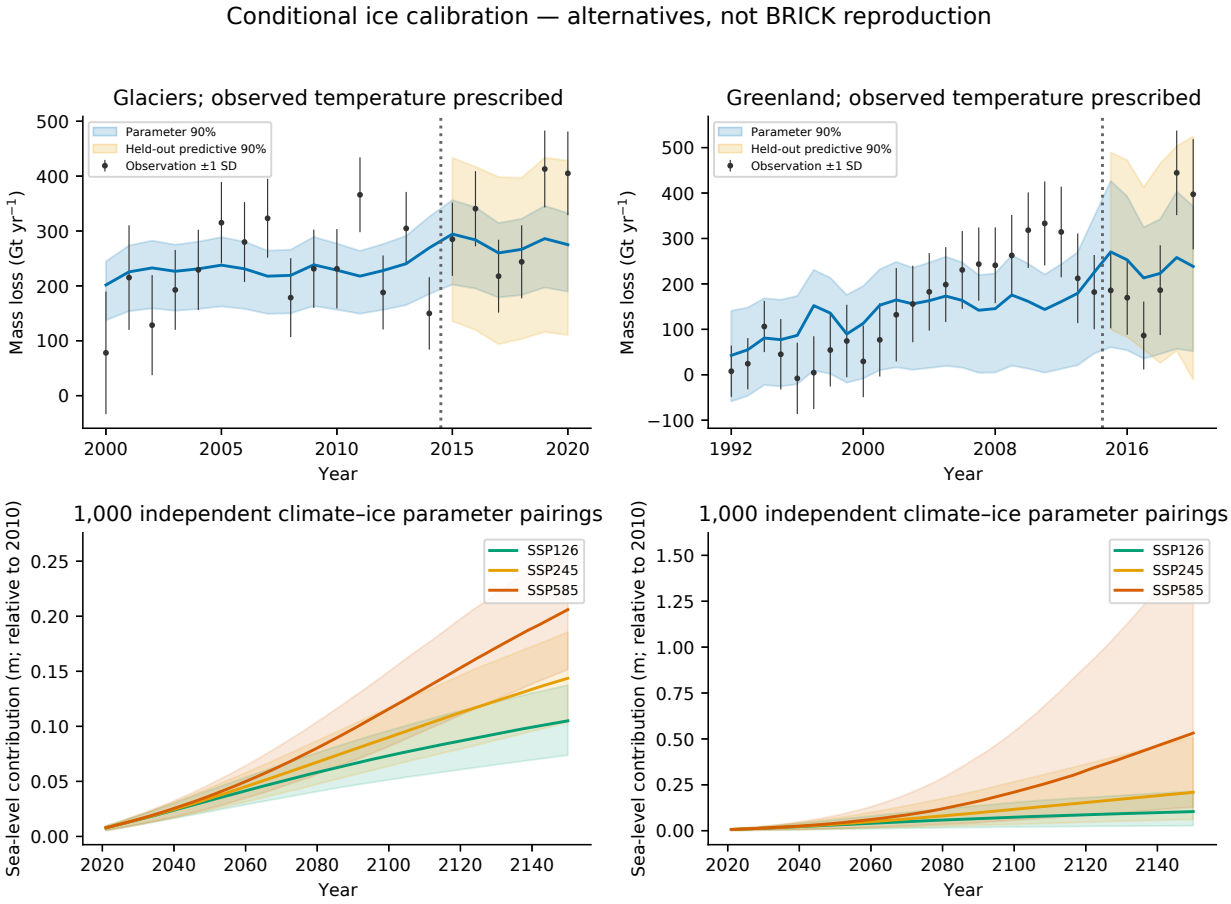


Figure 13: Conditional ice-component alternatives. Upper panels compare annual mass-loss rates with independent observations; the fit ends in 2014. Blue ribbons show parameter uncertainty, orange ribbons held-out predictive uncertainty including AR1 residuals and observational error. Lower panels show parameter-only projections relative to 2010 for three SSP forcing scenarios. These are separate component predictions, not a reproduced global sea-level distribution.

IMBIE gravimetry can contain peripheral glaciers also represented by GlaMBIE. Combining these series without an overlap correction could double count ice loss. The original component experiment therefore reports them separately. Otosaka and coauthors[19] quantify a4.1%/3.3% change in reconciled Greenland/Antarctic estimates when peripheral glaciers are removed from

gravimetry over2002–2020, below reported uncertainty. This is a specific accounting limitation, rather than a prohibition on any alternative total. An explicitly uncorrected component budget and covariance sensitivities can therefore be investigated, while exact target component boundaries and joint calibration remain unresolved.

8.3 Geographic exposure and regional accounting

SLIIDERS v1.1 supplies independently published elevation-binned land, population, capital, wetland, surge and socioeconomic inputs[21]. Only its input product was retrieved; no pyCIAM implementation, construction scripts or model outputs were consulted. Aggregating 11980 subadministrative records by coast and country yields 9574 groups. Six have no coastline and 37 lack positive economic inputs; their inventories and exclusion reasons remain in a separate CSV. The experiment uses the remaining 9531 coast-country groups.

The alternative uses SSP2 and SSP4 OECD economics in 2019 USD PPP, annual interpolation between decadal nodes, and a 2010–2100 horizon. Sea level is spatially uniform and prescribed as $h[(t-2010)/90]^{1.5}$, with $h = 0.5, 1, 2\,\mathrm{m}$. It has no temperature-conditioned probability interpretation. The model selects the least-cost first-period policy, retains its strategy and return level, and updates height or retreat thresholds in 2051. Costs are discounted to 2010 at 4%. Incremental cost subtracts a paired zero-SLR calculation. The lowest retreat level follows mean sea level, an explicit alternative to unavailable DINAS annual-high-tide inputs. Wetland exposure follows its actual elevation profile, with the documented rate-dependent loss fraction. These conventions do not recover the target's segmentation, monetary basis, decadal accounting, 2150 horizon or spatial fingerprints.

The initial pilot exposed a 9.0% quadrature discrepancy in no-adaptation costs: narrow surge densities could lie between nodes even when elevation breaks were resolved. Splitting integration at both elevation breaks and hazard quantiles reduces the maximum 12-versus-24-node difference to 1.5×10^{-11} across 114 pilot cases, with no policy changes. Original 0.1 m profiles additionally test the separate six-bin exposure approximation. Numerical integration agreement does not validate exposure coarsening. Indeed, original-resolution profiles change two of the 114 pilot policy choices and increase summed incremental cost by 9.6%. The largest individual relative change is 71.6%. This deterministic pilot is not representative sampling and cannot supply a global correction factor. The coarse global experiment remains a sensitivity result while a full original-resolution SSP4, 1 m calculation tests the global effect.

Regional ratios use independent World Bank national GDP[22]. We value 2010 GDP at 2019 PPP by multiplying 2019 current-PPP GDP by the country's 2010/2019 real-GDP ratio. Available landlocked-country GDP is included in denominators; coastal costs lacking a matching positive GDP are excluded from the ratios and their coverage is recorded. Afghanistan and Pakistan are restored to their pre-July-2025 South Asia classification, as documented by the World Bank. These explicit choices support an inequality sensitivity experiment, not reproduction of the target's regional numerical bars.

8.4 Complete original-resolution coastal comparison

The earlier19-unit pilot cannot establish a global elevation-resolution error. A subsequent paired experiment evaluates every eligible unit with all200 original0.1 m exposure bins, retaining the same inputs, prescribed trajectory, quadrature and policy conventions. Figure15 and the computed table report the complete comparison. Its cost difference is an effect of redistributing exposure within the original six bins, rather than adding population, capital or land to the domain. Independent binary and CSV readers agree exactly on the original-resolution pilot; doubled flood quadrature

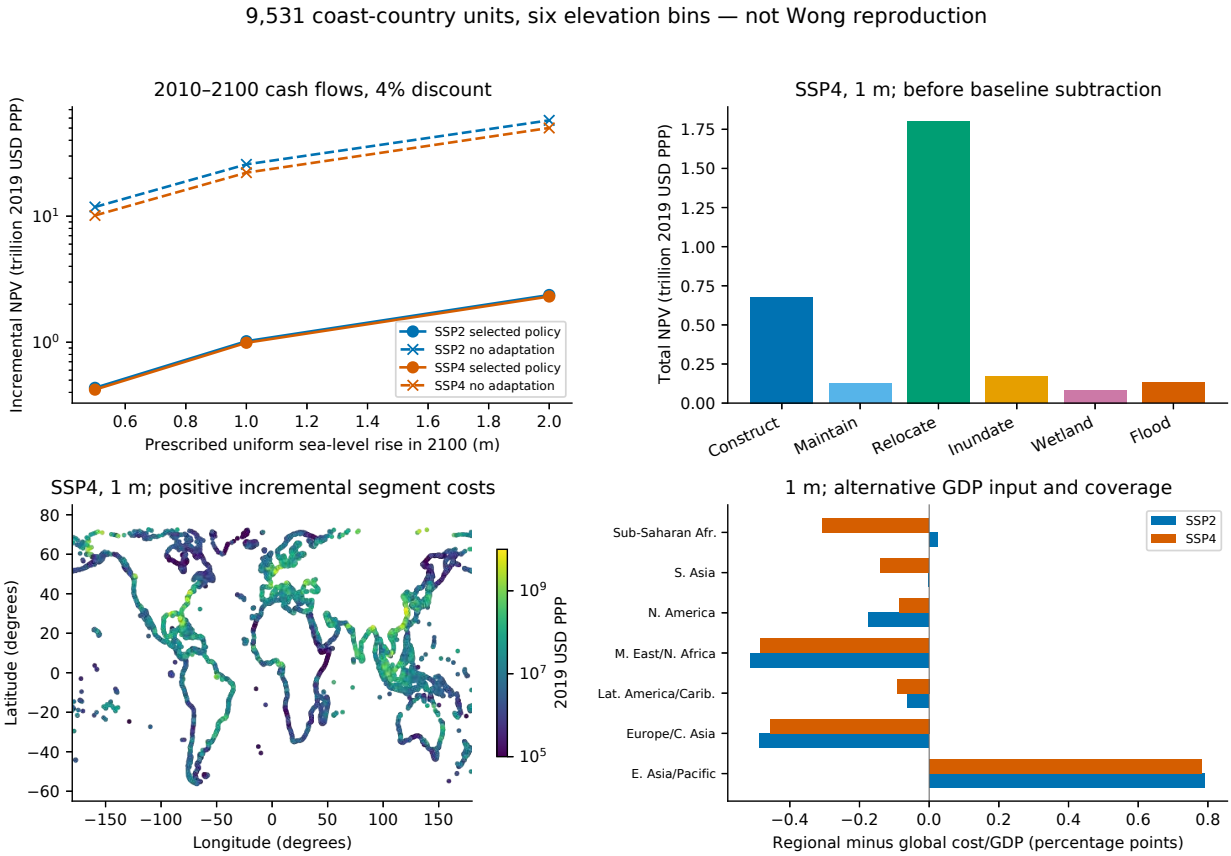


Figure 14: Geographic coastal alternatives. Upper left: global incremental costs for prescribed uniform sea-level paths, with paired no-SLR baselines for both selected policy and no adaptation. Upper right: unadjusted cost components for SSP4 and 1 m. Lower left: positive incremental costs at coast coordinates; points represent input coast-country groups, not population density. Lower right: regional minus global cost/GDP fractions at 1 m using independent GDP inputs and documented coverage exclusions. Every panel is a non-reproduction; the paths have no calibrated warming-bin or tail probability.

SSP	Rise (m)	Selected cost	No-adapt. cost	Migrants (million)
SSP2	0.5	0.435	11.824	99.465
SSP2	1.0	1.019	25.776	127.280
SSP2	2.0	2.370	57.592	197.987
SSP4	0.5	0.421	10.131	98.755
SSP4	1.0	0.990	22.141	125.951
SSP4	2.0	2.305	50.112	195.002

Table 5: Geographic alternative totals. Cost columns are trillion 2019 USD PPP, discounted to2010. Migrants count movements in the selected policy over2010–2100 and are not unique individuals over a demographic lifetime. No warming probabilities are assigned.

Ice component	Max $\widehat{R}$	Min bulk ESS	Min tail ESS	Holdout RMSE (Gt/yr)
Glaciers	1.0009	2752	1116	84.8
Greenland	1.0011	4449	2561	128.9

Table 6: Final conditional ice calibration diagnostics and held-out rate errors. Every coordinate and log density must pass $\widehat{R} < 1.01$ and both ESS$\geq$ 400. Observed temperature is supplied in held-out years.

changes costs by less than 2×10^{-13} in relative terms and changes no pilot policy. The preferred

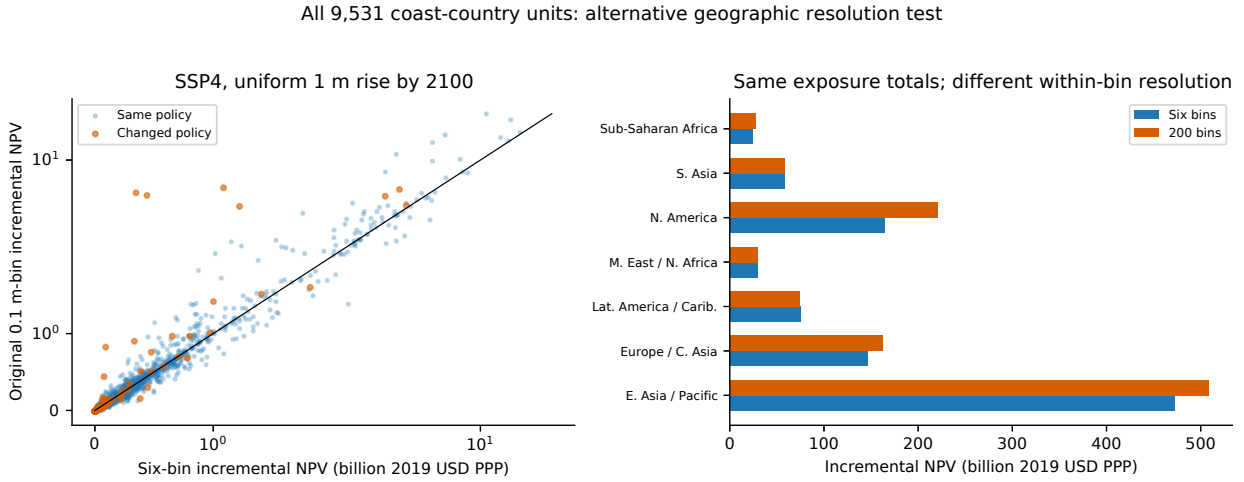


Figure 15: Matched SSP4,1 m coastal calculations for all9,531 eligible units. Left: incremental cost after subtraction of the paired zero-rise baseline; both axes use a symmetric logarithmic scale and the line marks equality. Right: World Bank regional totals under the same two elevation resolutions. Both calculations are geographic alternatives with uniform prescribed sea level, annual2010–2100 cash flows, and2019 PPP values. They are not recovered Wong warming-conditioned costs.

first-period policy changes in 148 units (1.55%). These totals replace extrapolation from the small coastal pilot; they do not establish convergence of the physical exposure data themselves.

9 Antarctic boundary recovery and conditional hindcasts

9.1 Recovered initial state and alternative forcing

The initial Antarctic state is specified more fully than the first assessment recognized. Shaffer's hindcasts begin at 240 kyr before CE 2000 with radius 1.8636×10^6 m and convert 24.78×10^{15} m^3 to 57 m of sea-level equivalent[7]. We now integrate this initial-value problem using independent proxy and instrumental inputs. NOAA Dome C and Spratt records use years before 1950; they are converted to calendar years before comparison with the model's different age origin. The W17F calibration dates of 118000,18000 and 4000 BCE correspond to 120,20 and 6 kyr before CE 2000[23].

Antarctic air temperature combines Dome C anomalies, the global land–ocean Mann EIV reconstruction, and HadCRUT4. We retain Shaffer's −0.1 K Dome C adjustment, independently rebase Mann to 1961–1990, and apply the documented modern amplification factor 1.2 around −18°C. The independently recomputed 500–1500 CE overlap adjustment would instead be −0.168 K;

Quantity	Six bins	200 bins	Change (%)
Selected policy total NPV	2.9932	3.0576	+2.15
Paired-baseline incremental NPV	0.9900	1.0992	+11.03
Movement count (million)	125.9506	128.5904	+2.10

Table 7: Complete matched resolution test for 9,531 coast-country units, SSP4 and prescribed1 m rise by2100. Cost rows use trillion 2019 USD PPP. The input elevation range is unchanged; only within-bin exposure resolution changes.

the reconstruction variant and datum therefore remain explicit uncertainties. The main alternative sea-level boundary subtracts Spratt's youngest 8.49 m value and connects its CE 950 point to the first available Church–White annual value at 1880.5. This interpolation is a declared assumption, not the recovered Lambeck splice. An unshifted Spratt case and both archived Waelbroeck chronologies test boundary sensitivity. These are observational proxy products, not recycled DAIS hindcasts[24, 25, 26, 27, 28].

The original transient subsurface-ocean history remains unrecovered. Shaffer's written steady-state approximation supplies a useful alternative:

$$T_o = 0.00690T_a^2 + 0.439T_a + 6.39 \quad [^\circ\mathrm{C}]. \tag{11}$$

Here T_a is Antarctic air temperature reduced to sea level. Applying this formula during a transient omits ocean thermal memory. It permits a test of the other equations without claiming recovery of the original DCESS forcing.

9.2 Volume integration and switching dynamics

The implementation advances physical ice volume by accumulation plus marine discharge and independently inverts the verified geometric relation to obtain radius at each integration stage. A separate radius-coordinate integration, including the prescribed sea-level derivative, agrees with the volume solution. All forcing knots and temperature gates split the integration grid; endpoint stages use the branch valid within that interval.

The additional fast-loss flux follows the recovered temperature threshold and stops below 18×10^{15} m^3[23]. Its cutoff creates a discontinuous differential equation. When slow accumulation would increase volume at the cutoff but full fast loss would decrease it, the mathematical sliding solution holds volume at the cutoff by balancing those fluxes. Slow mass loss can still lower volume below the cutoff. This continuation is derived and tested on exact piecewise trajectories; it is not an assertion about the original authors' timestep convention. Nineteen checks cover the geometry inversion, coordinate equivalence, constant forcing, hitting times, sliding and departure from the boundary.

For the central fixed-parameter hindcast, reducing the maximum step from 10 to 5 years changes saved sea-level equivalent by less than 8×10^{-8} m. That numerical agreement does not remove the physical mismatch: the predicted Last Interglacial contribution is -0.17 m, outside the written 3.6–7.4 m window. Figure16 compares input histories and their effects on this result. It also tests initialization and the sharply different responses obtained by changing the fast-loss threshold.

9.3 Conditional fitting and withheld annual rates

A separate experiment varies discharge exponent γ, ocean partition α, fast-loss rate λ and threshold T_c over their documented uniform ranges, fixing the remaining Shaffer parameters. Independent

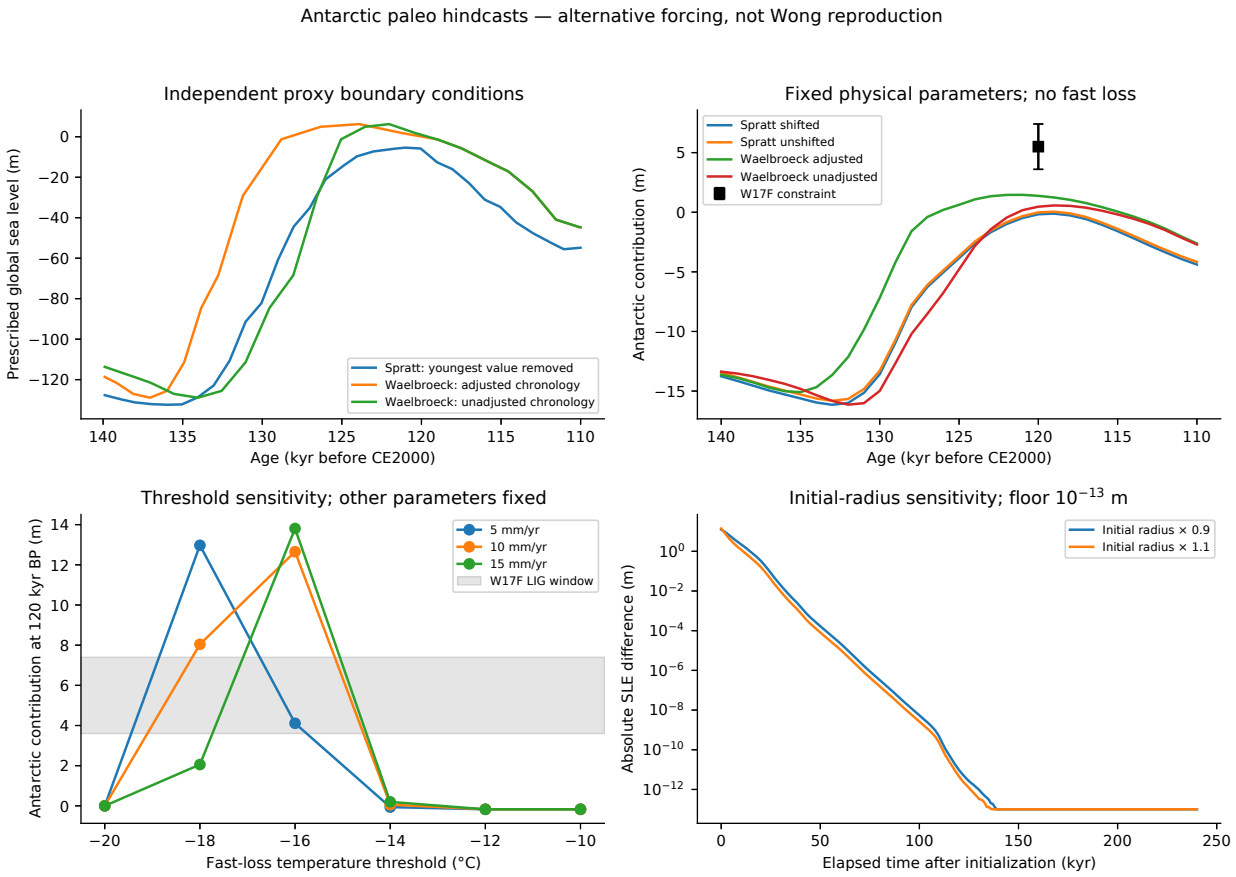


Figure 16: Antarctic hindcasts under explicit alternative forcing. Upper left: independent sea-level proxy inputs on a common calendar. Upper right: fixed Shaffer case 4 parameters without fast loss, with the W17F Last Interglacial constraint shown at 120 kyr before CE 2000. Lower left: fast-loss rate and threshold sensitivities with other parameters fixed. Lower right: differences after changing initial radius by 10%; the plotting floor denotes roundoff, not persistent physical differences. All panels are non-reproductions of Wong's calibrated Antarctic ensemble.

Latin hypercube draws assess parameter coverage before a bounded coordinate search minimizes the four standardized squared window residuals. The Last Interglacial hard range is also checked. This restricted point fit omits additional residual-process variance, uncertain forcing and the target joint calibration; it is neither a posterior nor an independent test of those four fitted values.

Annual IMBIE mass-loss observations after 2002 provide a separate conditional check. Observed temperatures remain prescribed in these years, and the calibration's modern cumulative constraint ends in 2002. Figure17 reports both parameter coverage and the later annual-rate errors. A good fit to four widely separated volume constraints need not predict short-period mass-loss variability.

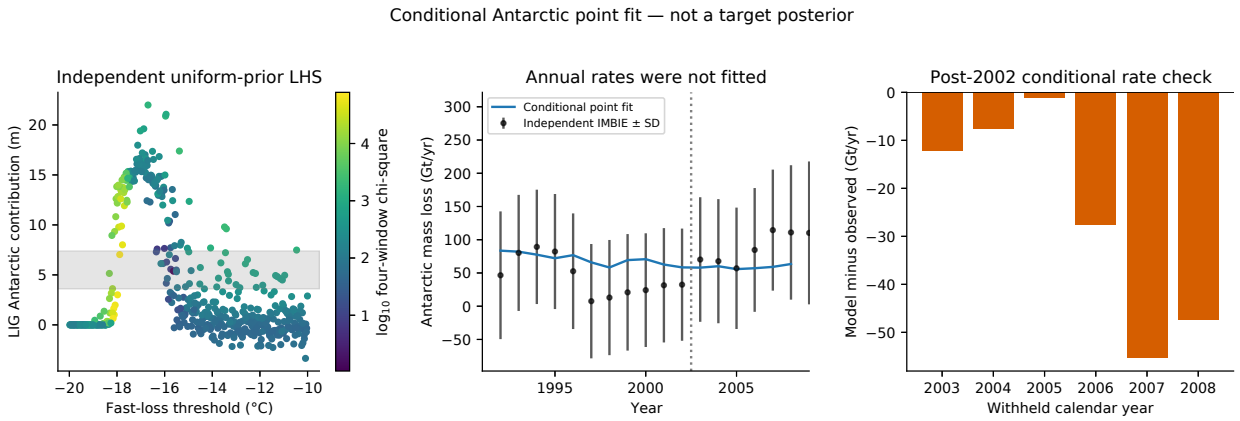


Figure 17: Conditional Antarctic fitting under alternative forcing. Left: independent uniform-prior Latin hypercube draws, coloured by four-window misfit; shading marks the Last Interglacial range. Centre: fitted-model annual mass loss and independent IMBIE observations with reported uncertainty. Right: model minus observed rates after the 2002 cumulative constraint. No posterior interval or recovered Wong result is implied.

Constraint	Written mean (m)	Written SD (m)	Point fit (m)
Last Interglacial	5.5	0.95	5.5
Last Glacial Maximum	-11.35	2.23	-11.392
Mid-Holocene	-2.63	0.68	-2.7276
1992–2002 increase	0.00197	0.00046	0.001951

Table 8: Conditional Antarctic predictions under the explicitly alternative forcing. These four quantities are fitted, not held-out validation. The Last Interglacial hard acceptance interval is 3.6–7.4 m.

The conditional solution uses $\gamma = 1.93747$, $\alpha = 0.02077$, $\lambda = 0.01161$ m yr^{-1} and $T_c = -15.62980\,^{\circ}$C. The search begins with 512 Latin hypercube points; 1 draws satisfy all four windows before local optimization. The final four-window χ^2 is 0.02267. Halving the maximum timestep from 10 to 5 years changes these predictions by at most 2.7e-09 m. The conditional 2003–2008 mass-loss RMSE is 32.35 Gt yr^{-1}, with bias -25.22 Gt yr^{-1}. The observed and predicted mean rates are 84.17 and 58.96 Gt yr^{-1}, respectively.

The earlier64-point pilot reaches a different solution, with fast-loss rate 0.015 m yr^{-1} and threshold$-15.484\,^{\circ}$C, while its four-window $\chi^2 = 0.02354$ is almost unchanged. The larger search gives a lower misfit without reaching that rate bound. These two fits illustrate conditional parameter ambiguity; neither is presented as a uniquely inferred fast-loss mechanism.

BRICK's written shore coupling is also recoverable: the Antarctic contribution has fingerprint−1, while the other contributions have factor+1[5]. Consequently, if S_o is the sum of other components and S_A is Antarctic global sea-level change, the local boundary changes as $S_o - S_A$. A volume integrator applies this feedback through geometric inversion at each stage. An independent radius-coordinate calculation includes its derivative in $\dot{S}$; ten checks cover that equivalence, equilibrium, timestep refinement and component accounting. A lagged-input alternative follows the documented previous-year forcing convention, while retaining an explicitly independent within-year numerical integrator.

10 Emissions probabilities and coupled future alternatives

10.1 Recovered priors and unresolved expert density

The written emissions constraints are more specific than the initial blocker assessment implied. Darnell's manuscript and its published successor specify the cumulative interval as 2022–2100 and describe annual selection of the nearest SSP CO_2 trajectory[2]. The prior peak year can occur after the simulation horizon. The implementation now retains such draws as growing trajectories through the last simulated year, rather than rejecting valid prior mass. Ten thousand independent draws contain eight such peaks after 2150. The starting net emissions, including the documented cement-carbonation adjustment to Global Carbon Budget 2023, are 40.663 $GtCO_2\ yr^{-1}$ in 2022.

Only 2.35% of these prior draws satisfy the 1223 $GtCO_2$ cumulative budget. The written 50% and 75% expert assessments therefore imply substantial reweighting. They do not specify a unique density, and the baseline requirement that SSP4-6.0 be modal imposes information beyond a single event probability. To isolate this ambiguity, two explicit alternatives minimize relative entropy with respect to the sampled uniform prior subject only to the budget-event probability. If A denotes draws within the budget, n_A its size and p the imposed probability, the weights are

$$w_i = \begin{cases} p/n_A, & i \in A, \\ (1-p)/(n-n_A), & i \notin A. \end{cases} \tag{12}$$

This formula follows from Lagrange multipliers and leaves conditional distributions within each class unchanged. It does not recover the elicited expert density or the SSP4-6.0 mode. The effective weighted sample sizes are 918 and 417 for $p = 0.5$ and 0.75. Changing peak-year rounding to floor or ceiling changes budget membership for 28 or 21 draws; trapezoidal rather than annual-mean summation changes 26 memberships. These sensitivities are retained explicitly.

10.2 Independent SSP inputs and carbon–climate coupling

The RCMIP protocol 5.1.0 supplies independent exogenous SSP emissions rather than Wong model outputs[29]. Total global CO_2, fossil and industrial CO_2, and land-use CO_2 are extracted for six SSPs and linearly interpolated between supplied annual nodes. Their partition identity closes to $2.9 \times 10^{-14}\ GtCO_2\ yr^{-1}$. No extrapolation is needed through 2150. The interpretation tested here selects, separately each year, the SSP with smallest absolute total-emissions difference. CEDA non-CO_2 forcing from that scenario is then linearly interpolated between annual nodes. A paired fixed-SSP2-4.5 calculation measures the consequence of this selection. The written nearest-trajectory statement alone does not settle all interpolation and distance conventions.

The future calculation starts from the conditional carbon hindcast and the accepted climate posterior's physical temperature state in 2021. Carbon feedback temperature is transformed to the

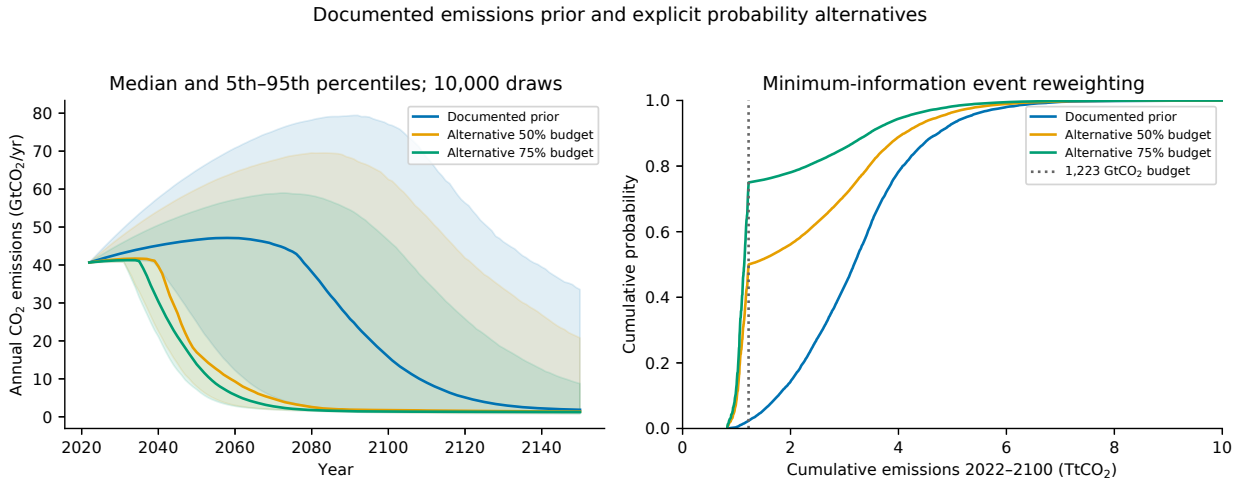


Figure 18: Documented emissions priors and explicit alternative weighting. Left: weighted medians and 5th–95th percentiles of annual emissions. Right: cumulative budget distributions and the 1223 $GtCO_2$ threshold. All distributions use the same 10,000 independent draws; only their weights differ. The two reweighted distributions are non-reproductions of the published emissions posterior.

same preindustrial datum used in the carbon fit. The future CO_2 forcing uses $3.7 \log_2(C/280)$ W m^{-2}; its difference from the historical CEDA forcing is reported as a restart discontinuity, without compensating adjustment. The carbon fertilization parameter is fixed at the independently fitted central value; uncertainty in this parameter and its dependence on climate parameters are omitted. The resulting distribution is therefore conditional and cannot stand in for the target joint posterior.

A balanced conditional sample contains 128 draws on each side of the budget. Complete accepted climate-posterior rows are paired independently with these emissions draws. Each probability measure retains its own class weights. An eight-member pilot with both forcing rules verifies carbon conservation and compares eight with sixteen coupling substeps per year. The maximum differences are 1.45×10^{-5} K in temperature and 3.96×10^{-4} ppm in CO_2; these checks precede the 256-member calculation. Warming categories use the annual 2100 temperature relative to 1850–1900, with boundary values assigned to the lower category exactly as in the publication.

Distribution	$T_{.05}$	$T_{.50}$	$T_{.95}$	$P_1(\%)$	$P_2(\%)$	$P_3(\%)$	$P_4(\%)$
Prior	1.136	1.746	2.650	26.0	63.3	10.7	0.0
50% alternative	0.811	1.291	2.552	61.7	32.8	5.5	0.0
75% alternative	0.784	1.117	2.211	80.5	16.8	2.7	0.0

Table 9: Alternative annual warming in 2100 relative to 1850–1900, in K. Probabilities refer to the four published threshold definitions. The balanced 256-member subsample has distinct weights for each probability measure; it is not the published 5,000-member ensemble.

The largest paired change in 2100 warming from nearest-scenario versus fixed-SSP2-4.5 non-CO_2 forcing is 0.378 K. The restart changes the CO_2 forcing by -0.1762 W m^{-2}, without compensation. Maximum carbon accounting error is 8.3e-12 Gt C.

No member of this256-member calculation exceeds3.5 K. The zero entry in the fourth category therefore denotes absence in this finite sample, not proof of zero probability. A separate tail-stratified emissions experiment is motivated by this limitation; it preserves empirical prior class masses with

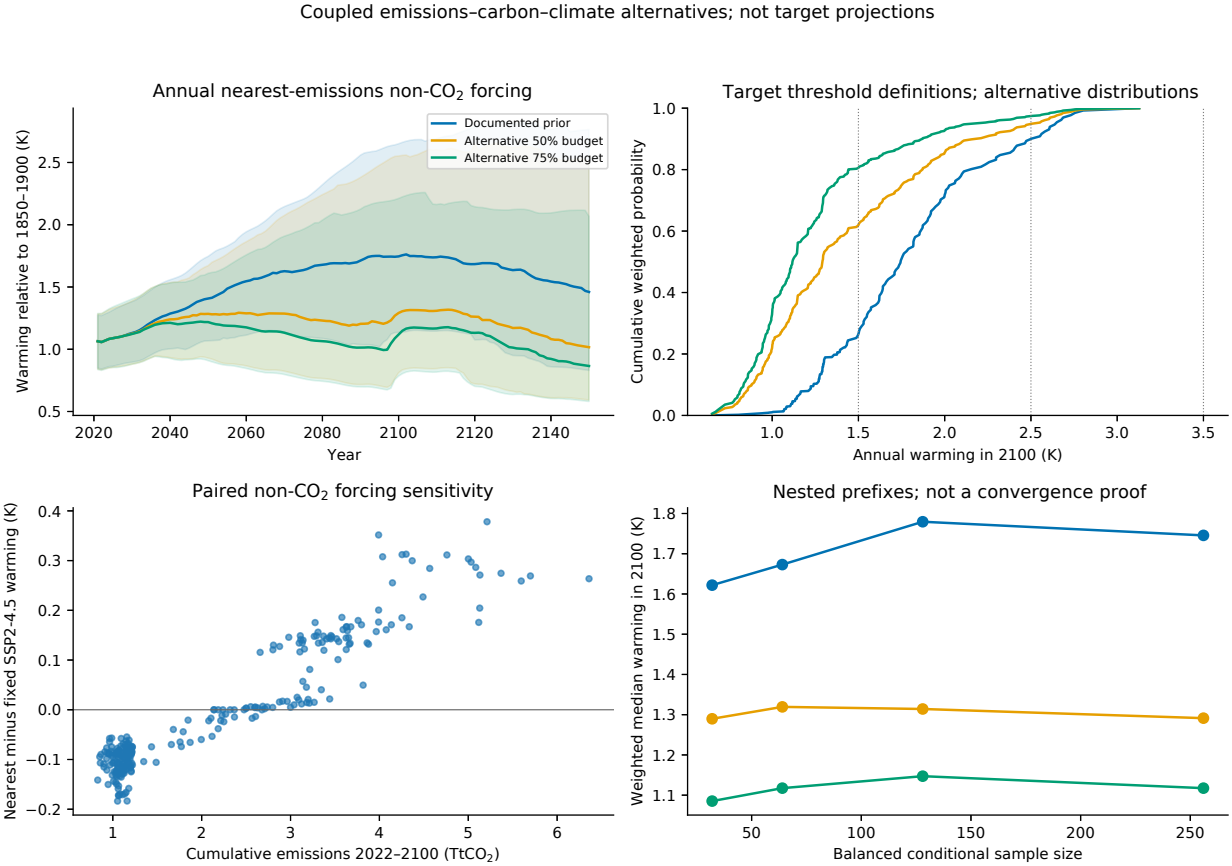


Figure 19: Coupled alternative future experiments. Upper left: median and 5th–95th temperature ranges for the three weighting measures. Upper right: weighted 2100 temperature distributions and the published category boundaries. Lower left: paired sensitivity to annual nearest-emissions versus fixed-SSP2-4.5 non-CO_2 forcing. Lower right: medians for nested balanced sample sizes; stability of these medians alone is not evidence of tail convergence. Carbon parameters are conditionally fixed and climate rows are independently paired, so no panel is a recovered published projection.

explicit weights rather than changing parameters to create high warming.

11 Unresolved inputs and implications

The joint parameter distribution is a central missing scientific object. The cited methodology combines paleo Antarctic constraints, modern climate and component-sea-level observations, uncertain emissions, and autoregressive residuals. Complete target likelihood definitions, data vintages, measurement errors, initial states, and residual treatment were not recovered. Reusing published posterior samples is prohibited here. Independent calibration is now being performed with versioned exogenous observations and explicitly alternative likelihood choices. Independent draws from older marginal quantiles would discard physically relevant dependence.

The historical DIVA exposure bundle is another independent obstacle. The CIAM supplement reports that its v1.5.5 database was no longer supported and that v1.8 was not public at that time [8]. The EEA provider directory currently reports unavailable software [11]. This does not prove that every DIVA dataset is inaccessible; it establishes that a complete permissible target bundle was not recovered in this investigation. The independent SLIIDERS segmentation is used only in labeled alternative experiments, not substituted as if it were the 12,148 target segments.

Independent GTSR data and six SSP forcing inputs have been acquired [12, 13]. The GTSR release supplies coastal features and a later release supplies Gumbel parameters, but invalid records, repeated station labels and return-level discrepancies require explicit treatment. The 2022 academic supplement was recovered in the scholarly preprint, resolving the six socioeconomic marginal priors and several decision conventions. Independent country economics and geographic exposure now support an alternative calculation. Original exposure, component fingerprints and the precise target input vintages remain unresolved.

Independent paleo records and the documented initial Antarctic radius have now been recovered. These are no longer categorical barriers to hindcasts. The remaining paleo uncertainty concerns the exact splice, chronology and transient ocean forcing, rather than absence of all boundary information. Emissions priors and the2022–2100 budget interval are also recovered; a unique expert density and its full joint calibration remain unspecified.

The active stages are therefore to extend the independent calibration, reconcile the remaining carbon/ice variants, and propagate the alternative coupled pathways through sea-level and geographic cost calculations. Small coupled experiments precede any target-sized ensemble. The regional alternatives do not establish quantitative agreement with target inequality.

12 Assessment of the research-agent process

The agent recovered much of the mathematical interface from distributed scholarly sources, distinguished versions of a mutable preprint, and implemented several independently checkable numerical components. The analytical tests separate solver accuracy from calibration and data availability. Cross-language flood integration supplies an additional check against transcription or implementation errors in the elevation primitive and GEV convention.

Two interpretation errors were avoided through source comparison. First, the older BRICK temperature-based expansion emulator is not the heat-based component described in the target lineage. Second, a later revision behind the same DOI cannot automatically be treated as the manuscript available before publication. Direct calculation then identified two tensions between printed equations and their prose descriptions. These checks do not reveal the original implementation.

The principal limitation is evidence recovery, not numerical throughput. An agent can write functioning code while failing to identify the scientific experiment if it supplies convenient values for missing inputs. This project does not count such code as a successful reproduction. The initial assessment stopped too early: further scholarly reading recovered major mechanisms and independent inputs. Observation-based calibration, carbon mass closure and DAIS integration are now demonstrated, while a global adaptation ensemble remains unreproduced. The failed holdout checks show why solver verification and posterior convergence cannot substitute for validation of the predicted observable.

The research log records source access failures, incidental search exposure, interpretation decisions, and verification outcomes. No subagents were used. The work followed Brad Lipovsky's supplied research instructions and academic style guide. This assessment concerns the recorded execution; it is not a general benchmark of all AI research systems.

13 Reproducibility and current scope

Run `bash scripts/run_all.sh` from the project root to execute the offline verification, generate numerical diagnostics and figures, and compile this report. Julia uses standard libraries only. Python 3.6.8, NumPy 1.19.5, SciPy 1.5.4 and Matplotlib 3.3.4 were available locally; pdfLaTeX comes from TeXLive 2018. The driver verifies executable availability and imports, requires no installation, and limits numerical libraries to one thread. Calculations run at scheduling priority 19 on a heavily loaded shared server. All artifacts, temporary files and tool caches are directed into the designated project. Source acquisition is a separate optional, hash-checked command. The scientific build does not require network access or downloaded authors' outputs.

The independent calculations verify response and cost mechanisms, quantify written-specification discrepancies, and test alternative observation-conditioned climate scenarios. An original conceptual Figure 4 is complete. Main Figures 1–3, supplementary Figures 1–10 and modeled Table 1 remain unreproduced. A printed-table arithmetic agreement is not promoted to numerical-model agreement. The missing calibration and coastal inputs must be resolved before an assessment of the target scientific conclusions can be claimed. These remaining tasks do not invalidate the completed mechanism tests or authorize treating this progress report as project completion.

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